

SOMATOM Scope

Maintenance Instructions System

Maintenance Instructions

This document is valid for:

SOMATOM Scope

SOMATOM Scope Power

The protocol C2-015.832.01.05.XX is required for these instructions

Document Version

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1.1 General Information



Only qualified and system-trained service staff is allowed to perform installation, service, maintenance, and quality assurance activities on the CT system.

Use only the most recent version of the service documentation.

Periodic, preventive maintenance on the system is mandatory to ensure continuous safety, reliability, and quality of the product.

Maintenance must be performed in accordance with these maintenance instructions.

Periodic maintenance includes safety inspection, preventive maintenance, preventive parts replacement, and quality assurance.

To indicate the type of maintenance task, the check points are specified and marked with attributes in the instructions as well as in the protocol.

The attributes are as follows:

Type of maintenance task	Explanation
PM	Preventive Maintenance
SI	Safety Inspection (in accordance with DIN EN 62353, IEC 62353)
SIM	Mechanical Safety Inspection
SIE	Electrical Safety Inspection
Q	System Quality

The results of the maintenance steps must be documented in the maintenance protocol once they have been performed.

Depending on country-specific specifications, either a paper version or an electronic version (forms4mobile on a handheld/PDA) of the protocol is used for documenting the maintenance results.

Explanation:

Because the new workflow-optimized full maintenance procedure contains a mixture of annual and semi-annual steps, an interval identification code has been implemented at the end of each PM/SI maintenance checkpoint text to easily identify the interval of each step.

For example: "PM - Gantry: Main bearing lubricated (12M)"

- (6M) = semi-annual step (every 6 months)
- (12M) = annual step (every 12 months)
- (24M) = biennial step (every 24 months)
- (30M) = two and a half year steps (30 months)

1.2 Maintenance Types - Overview

The recommended maintenance work for the system is divided into the following maintenance types and intervals:

Maintenance type	Maintenance interval	Estimated time
Partial maintenance ^{see note}	Every 6 months	2.5 hours
Full maintenance	Every 12 months	5 hours
Preventive maintenance	Every 2 and a half years	0.5 hours
Annual maintenance activities:		About 7.5 - 8 hours (without options)

The estimated time reflects an average value for the total efforts of each maintenance type and for a standard system configuration.

Maintenance efforts for optional components need to be added to the total time (as applicable).

Note:



Reduced maintenance opportunity for low-usage customer sites (< 15 patients /day)

Do not perform partial maintenance if the regional service organization has negotiated a reduced maintenance offering with a low-usage customer (patient throughput < 15 patients/day). In this case only the annual full maintenance procedure is required. It is the responsibility of the regional service organization to monitor the actual system usage and relevance of reduced maintenance for said system.

1.3 Partial Maintenance - Overview

1.3.1 Maintenance Interval & Time

Maintenance type	Maintenance interval	Estimated time
Partial maintenance	Every 6 months	about 2.5 hours (without options)

1.3.2 Maintenance Steps & Time

Component	Maintenance step/activity	Estimated time
System	System switch-on	5 min.
System	System switch off	2 min.
Gantry	Cover removal	10 min.
PHS1A	Cover removal	18 min.
PHS1A	Belt tension check (The drive belt tension should be checked only once after the first PHS installation, replacing the tabletop drive belt, horizontal drive and toothed belt.) PM:Drive belt tension check (6M)	15 min.
Mobile kit (option)	Inspections for systems with mobile kit PM:Inspection for damages (6M) PM:Inspection for damages and loose parts (6M) PM:Inspection for damages, loose parts, cracks/tears, screw connections (6M)	60 min. ¹
Sliding gantry (option)	Check the switching elements PM - check switching elements, front and rear (6M)	5 min. ²
Gantry	Slip ring cleaning PM:Slip ring assembly cleaned (6M)	10 min.
Gantry	Power brush (565V DC) cleaning / check / replacement PM: Power carbon brushes (565V DC) checked (6M) PM: Power carbon brushes (565V DC) replaced (6M)	5 min.

Component	Maintenance step/activity	Estimated time
Gantry	Power brush (230V AC) cleaning / check / replacement PM: Power carbon brushes (230V AC) checked (6M) PM: Power carbon brushes (230V AC) replaced (6M)	5 min.
Gantry	Signal brush block cleaning / check / replacement PM: Signal brush block checked (6M) PM: Signal brush block replaced (6M)	5 min
Gantry	MAS Air filter check / replacement PM:Air filter checked (6M) PM:Air filter replaced (6M)	5 min.
IMS	ICS and IRS cleaning PM:ICS and IRS cleaned (6M)	15 min.
System	Reinstalling covers SIM:Covers and ground wire cables installed (6M)	20 min.
System	System cleaning SIM:Flat screen cleaned (6M)	5 min.
System	System switch-on	5 min.
System	Image quality check Q:Quality check completed (6M)	25 min.
System	System availability check Q:Condition of CT system checked (6M)	5 min.
Total estimated time (without options)		155 min. = 2.5 hours (approx)

1. Only required if option is installed
2. Only required if option is installed

1.3.3 Required Tools & Measurement Equipment

Description	Part number	Location/usage
Standard tool kit	Local supply	General service work

Description	Part number	Location/usage
Quality phantoms	Delivered with the system	Image quality
caliper	Local supply	Gantry
Vacuum cleaner	Local supply	Gantry, IMS cleaning
Spring scale	Local supply	PHS1A belt tension
Ruler	Local supply	PHS
Soft cloth or lint-free towel	Local supply	Gantry, PHS, IMS
Cleaning agent	Local supply	IMS

1.3.4 Required consumable Items & Materials

Description	Location/usage	Part number (contains)	Amount required
Power brush kit	Gantry	03818254	1
Signal brush kit	Gantry	03804692	1
Air filter for MAS ventilator	Gantry	03068384	1

1.4 Full Maintenance - Overview

1.4.1 Maintenance Interval & Time

Maintenance type	Maintenance interval	Estimated time
Full maintenance	Every 12 months	about 5 hours (without options)

1.4.2 Maintenance Steps & Time

Component	Maintenance step/activity	Estimated time
System	System switch-on	5 min.
System	System switch off	2 min.
Gantry	Cover removal	10 min.
PHS1A	Cover removal	18 min.
System	Visual Inspection according to DIN EN 62353/IEC 62353 SI:No safety-relevant damage (12M) - Gantry SI: No safety-relevant damage (12M) - PHS SI:No safety-relevant damage (12M) - System cabinets SI:No safety-relevant damage (12M) - Care Vision (option) SI:Cables and cable routing indicate no visible damage (12M) - System SI:Cables and ducts routed safely, no visible damage (12M) - Care Vision (option) SI:In good shape, without visible damage (12M) - PHS pads SI:In good shape, without visible damage (12M) - Phantoms & Holder SI:In good shape, without visible damage (12M)- Osteo Accessory (option) SI:In good shape, without visible damage (12M)- Foot switch (option) SI:In good shape, without visible damage (12M) - Joystick (option)	10 min.

Component	Maintenance step/activity	Estimated time
System	Inspections for systems with mobile kit SI:Inspection for damages (12M) SI:Inspect for damages and loose parts (12M) SI:Inspect for damages, loose parts, cracks/tears, screw connections (12M)	60 min. ¹
System	Label inspection SI:Labels inspected (12M)	2 min.
PHS1A	Vertical drive spindle lubrication PM:Vertical drive spindle lubricated (12M)	2 min.
PHS1A	Guide rail lubrication PM:Guide rails lubricated (12M)	3 min.
PHS1A	Horizontal belt tension check (The drive belt tension should be checked only once after the first PHS installation, replacing the tabletop drive belt, horizontal drive and toothed belt.) PM:Belt tension check (12M)	15 min.
PHS1A	Anchoring check PM: Anchoring check (12M)	10 min.
MCS (option)	Visual and function check SI:Visual checks for varnish damage (12M) SI:Visual checks for cracks (12M) PM:Horizontal and vertical movements checked (12M) PM:Mechanical end stops checked (12M) PM:Load balance checked (12M)	15 min. ²
Sliding gantry (option)	Check sliding gantry PM - check toothed belt (12M) PM - check switching elements, front and rear (12M) PM - check protective conductor connection (12M)	15 min. ³
Gantry	Drive belt tension check PM:Drive belt tension check (12M)	5 min.
Gantry	Rotation brake check PM:Gantry brake checking (12M)	5 min.

Component	Maintenance step/activity	Estimated time
Gantry	Cover hooks check PM:Checking the elasticity of the hook holders on both the front and rear of the gantry (12M)	1 min.
Gantry	Gantry air outlets cleaning PM:Air outlets cleaned (12M)	5 min.
Gantry	Gantry cooling fan check PM:Gantry cooling fans checked (12M) PM:Gantry cooling fans checked (12M)	2 min.
Gantry	Slip ring cleaning PM:Slip ring assembly cleaned (12M)	5 min.
Gantry	Power brush (565V DC) cleaning / check / replacement PM: Power carbon brushes (565V DC) checked (12M) PM: Power carbon brushes (565V DC) replaced (12M)	5 min.
Gantry	Power brush (230V AC) cleaning / check / replacement PM: Power carbon brushes (230V AC) checked (12M) PM: Power carbon brushes (230V AC) replaced (12M)	5 min.
Gantry	Signal brush block cleaning / check / replacement PM: Signal brush block checked (12M) PM: Signal brush block replaced (12M)	5 min.
Gantry	Air filter check / replacement PM:Air filter checked (12M) PM:Air filter replaced (12M)	15 min.
Gantry	Main bearing lubrication PM:Main bearing lubricated (12M)	10 min.
Gantry	MAS air filter check / replacement PM:Air filter checked (12M) PM:Air filter replaced (12M)	5 min.
Gantry	Tube cooler cleaning PM:Tube cooler cleaned (12M)	10 min.
IMS	PSB/UPS checking and ICS/IRS cleaning PM: PSB/UPS position checked (12M) PM:ICS and IRS cleaned (12M)	15 min.

Component	Maintenance step/activity	Estimated time
System	System protective conductor checked SIE:LCB: Protective conductor resistance $\leq 100 \text{ m}\Omega$ (12M) SIE:Gantry: Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M) SIE:Patient table: Protective conductor resistance $\leq 300 \text{ }\Omega$ (12M) SIE:IRS Tower PC: Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M) SIE:ICS monitor: Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M) SIE:PSB/UPS for ICS: Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M) SIE:IRS Tower PC: Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M) SIE:Ceiling support (in scan room, option): Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M) SIE:Monitor trolley (in scan room, option): Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M) SIE:Monitor (in scan room, option): Protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)	20 min.
System	System switch-on	5 min.
System	Function Test SI:Emergency STOP circuit checked (12M) SI:Radiation cut-off checked (12M) SI:Radiation indicators checked (12M) SI:Door contact switch checked (12M) SI:Patient positioning aids checked (12M) SI:Function of the Line Connection Box checked (12M)	20 min.
Gantry	Gantry tilting check (12M)	10 min.
System	Cover reinstallation SIM:Covers and ground wire cables installed (12M)	20 min.
System	Image quality check Q:Quality check completed (12M)	25 min.
System	System cleaning SIM:Flat screen cleaned (12M)	5 min.

Component	Maintenance step/activity	Estimated time
System	System function check Q:Condition of CT system checked (12M)	5 min.
Total estimated time (without options)		295 min. = 5 hours (approx.)

1. Only required if option is installed
2. Only required if option is installed
3. Only required if option is installed

1.4.3 Required Tools & Measurement Equipment

Description	Part number	Location/usage
Standard tool kit	Local supply	General service work
Quality phantoms	Delivered with the system	Image quality
Torque wrench 25 - 130 Nm	03424561	PHS
caliper	Local supply	Gantry
Grease gun with NCA52	Local supply	Gantry/PHS
Grease gun with Tungaloy B.S.	Local supply	PHS
Ground wire meter	4415899 RV090	System
Vacuum cleaner	Local supply	Gantry, IMS cleaning
Spring scale	Local supply	PHS1A belt tension
Feeler gauge	Local supply	Gantry belt tension
Ruler	Local supply	PHS1A
Soft cloth or lint-free towel	Local supply	Gantry, PHS, IMS
Cleaning agent	Local supply	IMS

1.4.4 Required consumable Items & Materials

Description	Location/usage	Part number (contains)	Amount required
Power brush kit	Gantry	03818254	1
Signal brush kit	Gantry	03804692	1

Description	Location/usage	Part number (contains)	Amount required
Isoflex Topas NCA 52	Gantry/PHS	04007725 (400g package)	as required
Tungrease B.S.	PHS1A	03473287 (2.20lb package)	as required
Filter at the bottom of the gantry	Gantry	10355963	1
Filter at the left side of the gantry	Gantry	10355961	1
Filter at the back of the gantry	Gantry	10355962	1
Air filter for MAS ventilator	Gantry	03068384	1

1.5 Preventive Parts Replacement - Overview

HenDan1500VA UPS (P/N:10049948) have to be replaced every 2.5 years beginning with the date of installation. The date for the next replacement has to be documented in the Maintenance Protocol.

1.5.1 Replacement Interval and Time

Component	Maintenance steps	Replacement interval	Estimated time
IMS	UPS Replacement	Every 2.5 years	30 min.
Total estimated time:			30 min.

1.5.2 Maintenance Steps & Time

Component	Maintenance step/activity	Estimated time
System	System switch off	2 min.
IMS	UPS Replacement (30M)	18 min.
System	System switch-on	5 min.
System	System function check Q:Condition of CT system checked (30M)	5 min.
Total estimated time (without options)		30 min.

1.5.3 Required Tools & Measurement Equipment

Description	Part number	Location/usage
Standard tool kit	Local supply	General service work

1.5.4 Required consumable Items & Materials

Description	Location/usage	Part number (contains)	Amount required
HenDan 1500VA UPS	ICS, IRS	10049948	1

1.6 Maintenance Documentation

A maintenance protocol must be generated after every maintenance activity.

The following protocol belongs to these maintenance instructions:

Document	Print number
Maintenance protocol	C2-015.832.02.xx.02 Note: The maintenance protocol must have the same document version (xx) as the maintenance instructions.

The completed protocol must be forwarded to the customer (as a printout or electronic copy).

The archiving of the maintenance protocols near the system is the customer's/operator's responsibility. We recommend archiving the protocol in the System Owner Manual.

We also recommend that the country's organization keep a copy of the maintenance protocols to be able to reconstruct the maintenance history to prove proper system condition and operation (e.g. in case an incident/damage should occur).

1.6.1 Rules for filling out the Maintenance Protocol

- A protocol must be generated every time a maintenance activity is performed.
- All entry fields in the protocol must be completed.
- The checkbox **"OK"** must be marked for each successfully performed work step or test.
- The checkbox **"not OK"** must be marked for each work step or test that was not performed successfully (error occurred, value out of tolerance, etc.).
- The checkbox **"n.a."** must be marked for work steps or tests that cannot be performed (for non-installed options, etc.).
- Mark any entry fields that do not apply to the system configuration (option not installed, etc.) with **"n.a."**. Also, if there is no information available for an entry field, enter **"n.a."**
- **NOTE:** the date must be entered in the following format: **yyyy-mm-dd**, except when especially specified.

Explanation: yyyy=year (2012...); mm=month (01...12); dd=day (01...31)

1.7 Safety Information

1.7.1 Safety inspection

Safety inspection is required every 12 months.

All safety inspection steps are integrated into the annual "Full maintenance" procedure of the system.

1.7.1.1 Measuring the Leakage Currents

- Device leakage current:

- » not required

Reason: Hard-wired connection and on-site protective measures in case of indirect contact (e.g., in accordance with DIN VDE 0107).

- Patient leakage current at the application section of the patient table (type B):

- » not required

Reason: Insulation between parts under voltage and application section, though metal parts with ground wire connections.

- Patient leakage current in the ECG module integrated into the table (type BF):

- » not required

Reason: Design measures and fixed installation.

1.7.1.2 Measuring the Insulation Resistance

- » not required

Reason: Not necessary since the system is a Class A system and the visual check was completed successfully.

1.7.2 Precautions for working in the gantry

Only qualified persons trained on the systems are allowed to perform CT system installation, service, maintenance and quality assurance.

To avoid any risk of injury to persons or damage to the system, the safety instructions must be read and observed in the:

General Safety Notes (TD00-000.860.01.xx.xx)

and

Product-specific Safety Notes (C2-015.860.01.xx.xx)

The product-specific safety notes contains important information about:

- General safety information about the gantry, patient table, and laser products.
- Attachment of screws and nuts, adjustment instructions for torque wrench and use of Loctite.
- Safety advices (Hazard Keys)

1.7.2.1 High voltage



Dangerous voltages may be present.

Switch off the main power supply before working on the system and secure the system against unintended switch-on.

Allow at least for two minutes discharge time after switching off the system. Prior to starting work on the components indicated below, measure the voltage level in the PDS between fuse holder F3.2(+) and F4.2(-) . The voltage must be below 40 V DC.

1.7.2.2 Working in the Gantry

- Prior to starting work in the gantry:
 - Wait until rotation has stopped
 - Disable gantry rotation via service switch S5 on MAS D301
 - Use the mechanical brake to prevent gantry rotation while working in the gantry.
 - Connect the stationary and rotating part of the gantry with a ground cable before working in the gantry.



Remove the cable prior to rotating the gantry.

1.7.2.3 Automatic Procedures



Do not leave the system unattended when a program is running with automatic movements or automatic radiation.

1.7.2.4 Radiation Safety



Observe radiation protection regulations.

1.7.2.5 Bloodborne pathogens



Take appropriate precautions against exposure to blood-borne pathogens when handling parts of the system that may have come into contact with patients.

1.8 Details of the consumable Carbon Brushes

1.8.1 Power Carbon Brush Kit

Tab. 1 Content of carbon brush set (power)

Contents	Item number	Number	Required
Carbon brush set (power)	03818254	1	1
▪ Carbon brushes CA40 (6 x 6 x 20mm)		19	19
▪ Screw caps (M10 x 1)		2	Spare

1.8.2 Signal Carbon Brush Kit

Tab. 2 Content of carbon brush set (signal)

Contents	Item number	Number	Required
Carbon brush set (signal)	03804692	1	1
▪ Signal brush block		1	1
▪ Carbon brushes CA40 (3 x 4 x 20 mm)		8	8
▪ Screw caps (M7 x 0.75)		2	Spare

1.9 End of General Information

End of chapter

- » Continue with the **“Partial Maintenance”** or **“Full Maintenance”** chapters, depending on the maintenance type to be performed.

Maintenance type	Maintenance interval	Total estimated time:
Partial maintenance	Every 6 months	150 min.

2.1 Switching on the System

Required time:	5 min.
Required tools	n.a.
Required materials	n.a.

If the system is not in standby already, switch on the system.

- Switch on the LCB.
- Switch on the PSB/UPS for the IMS.
- When the system reaches the standby mode, elevate the tabletop to its uppermost position and move it fully into the gantry.

2.2 Switching off the System

Required time:	2 min.
Required tools	n.a.
Required materials	n.a.

Shut down the computers and power off the whole system and secure it against unintended switch on.

- In the ICS main user menu, select **System > End > Shutdown System** to shut down the computers.
- Switch off the power supply to the entire system and secure it against unintended switch on.

2.3 Gantry: Cover removal

Required time:	10 min.
Required tools	Standard tool kit
Required materials	n.a.



Read and observe the safety information contained in the "Safety information" section of this document.



Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained.

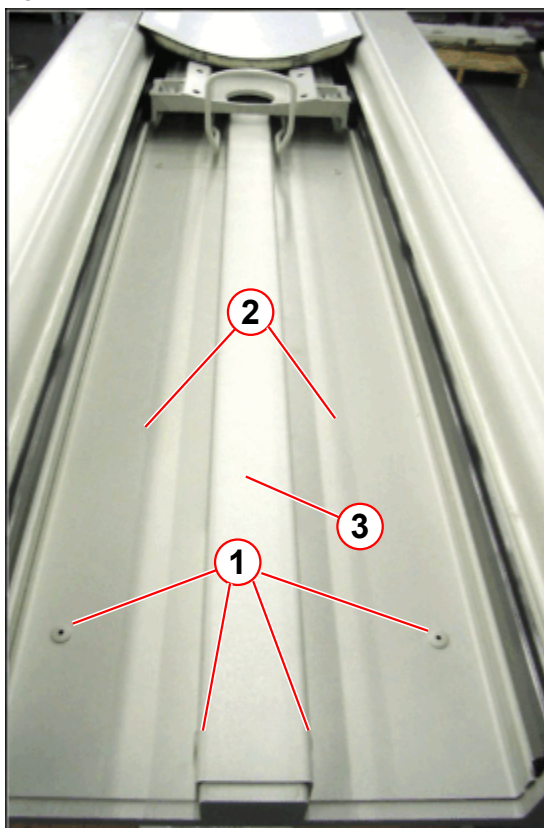
- Release the tabletop and manually move it to its Home position.
- Remove the rot front and rot back assembly.
- Remove the filter assembly.
- Remove the right stand in back and the right stand of the main assembly.

2.4 PHS1A(P/N:10355389): Cover removal

Required time:	25 min.
Required tools	Standard tool kit
Required materials	n.a.

- Remove the covers of PHS1A (P/N 10355389) as follows:

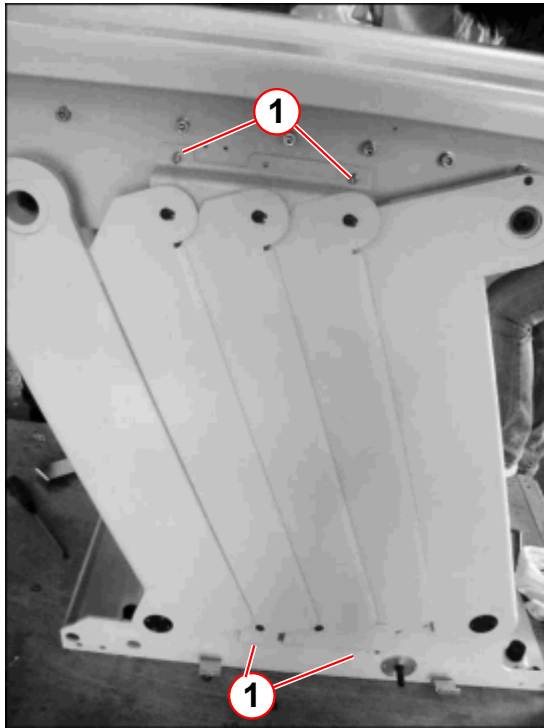
Fig. 1: Remove two back-end metal sheets



- (1) White screws
- (2) Metal sheets at the back-end
- (3) Metal bracket of the belt

- Remove two back metal sheets (pos.2) so that 6 lubricating nipples of the tabletop are accessible from underneath after the tabletop is moved to the back position.

Fig. 2: Cover plates



(1) 4 x screws

- Remove the back basin and center cover
- Remove the bottom cover.
- Remove the cover plate assembly with 4 screws (pos.1) so that the lubricating nipple of the vertical drive is accessible.

2.5 Mobile Kit (Option): Inspections

Required time:	60 min.
Required tools	Standard tool kit; torque wrench 25 - 130 Nm
Required materials	n.a.

The mobile CT requires additional maintenance due to extraordinary conditions.

2.5.1 Transportation Lock Inspection

PM Inspection for damages (6M)

- Visually inspect for damages and test their functions.

2.5.2 Table Inspection

PM Inspection for damages and loose parts (6M)

- Visually inspect for damage, loose screws and parts as well as cracks/tears, especially at stressed parts.
- To check attachment of the PHS to adequate flooring, please refer to the guidelines of the trailer manufacturer (refer to references in the system-specific Planning Guide C2-025.891.01.xx.02).

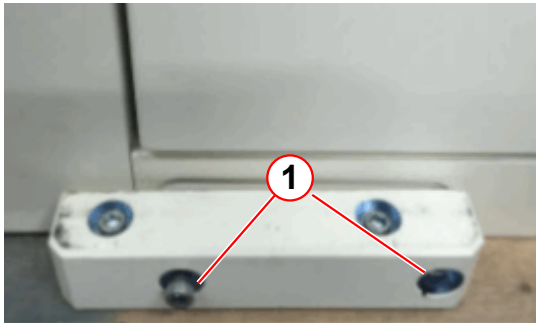
2.5.3 Gantry Inspection

PM Inspection for damages, loose parts, cracks/tears, screw connections (6M)

- Visually inspect for:
 - Damages
 - Loose screws and parts
 - Cracks / tears especially at stressed parts and along visible welding seams, particularly at the gantry base

- Inspect the following using a torque wrench:

Fig. 3: Block at gantry stand



(1) Screws

The screw connection between each one of the 4 blocks and the gantry (torque 90 Nm)

Fig. 4: Pedestal

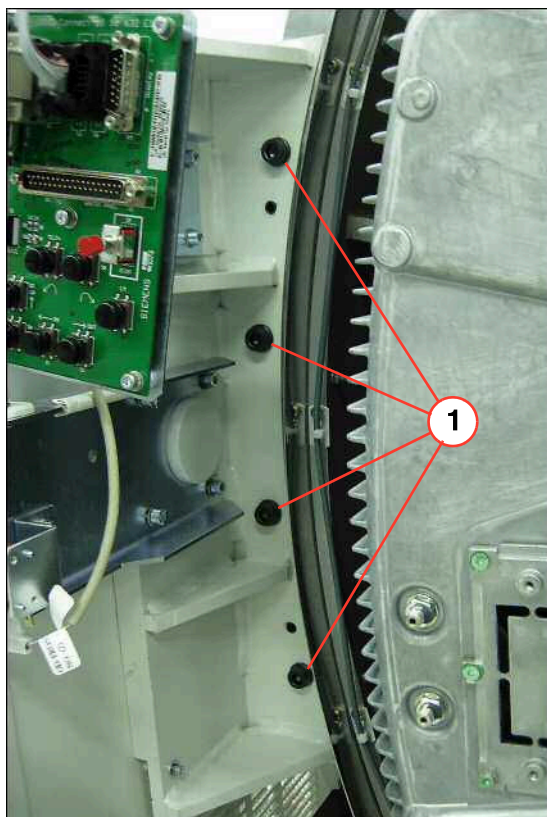


(1) Screws

The left and right-hand side pedestal (torque 50 Nm) - 2 screws (pos.1) on each side

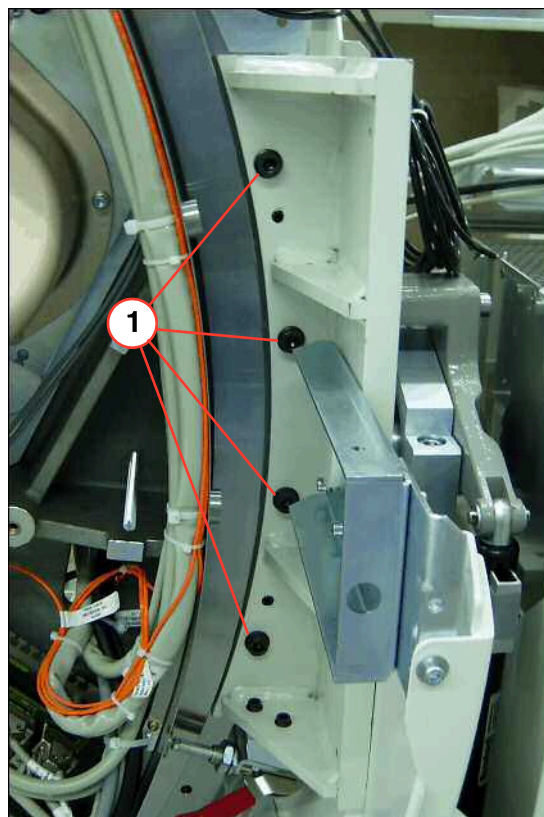
- The left and right hand side tilt suspension (torque 90 Nm) - 4 screws (pos.1) on each side

Fig. 5: Tilt suspension (left hand side)



(1) Screws

Fig. 6: Tilt suspension (right hand side)



(1) Screws

- The X-ray tube (torque 90 Nm)
- The high voltage generator (torque 90 Nm)
- To check the attachment of the gantry to adequate flooring, please refer to the guidelines of the trailer manufacturer (refer to the reference in the system-specific Planning Guide C2-015.891.01.xx.02).
- If necessary, vacuum clean the cover to remove synthetic particles produced by the cover rubbing against the collimator during transportation.

2.6 PHS1A(P/N:10355389): Checking the Drive Belt Tension

PM Drive belt tension check (6M)

Required time:	15 min.
Required tools	Standard tool kit; ruler; spring scale
Required materials	n.a.

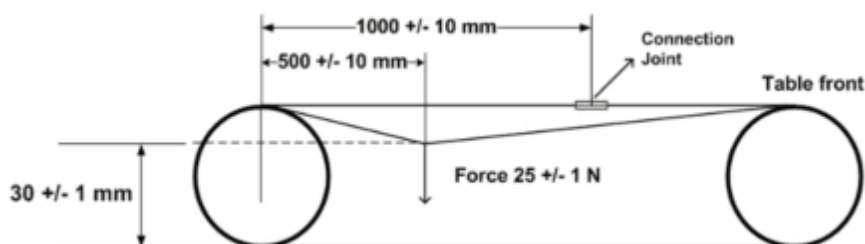


The drive belt tension should be checked only once after the first PHS installation, replacing the tabletop drive belt, horizontal drive and toothed belt.

Check the horizontal drive belt tension:

- Move the table forward 1000 +/- 10 mm so that the distance between the tension adjuster and the connection joint is 1000 +/- 10 mm.
- Loosely install a cable tie around the belt and position it approx. in the center of the tension adjuster and the connection joint ($1000\text{mm}/2 = 500\text{mm}$).
- Hook a spring scale (100 N) into the cable tie.
- Position a ruler or measuring tape in the vertical position next to the cable tie.
- Pull the spring scale in the vertical direction with a tension of 25 +/- 1 N and measure the vertical change in distance (see below illustration).
 - » The vertical change in deflection must be 30, +/-1 mm (at 25, +/- 1 N).
- Adjust the belt tension accordingly with the adjustment screw until it is within the tolerance margin.

Fig. 7: Belt tension



2.7 Sliding gantry (option): Check the switching elements

PM Check switching elements, front and rear (6M)

Required time:	5 min.
Required tools	n.a.
Required materials	n.a.

- Check the switching elements by moving the gantry in the forward and reverse directions.
 - » Push your foot against the switching element to check the stop function.

2.8 Gantry: Cleaning the Slip Rings

PM Slip ring assembly cleaned (6M)

Required time:	10 min.
Required tools	Vacuum cleaner
Required materials	n.a.

- Remove the carbon dust from the slip ring using a vacuum cleaner while rotating the gantry step-by-step counter-clockwise by hand.



Do not touch the surface of the slip ring with your fingers or with parts of the vacuum cleaner.

2.9 Gantry: Checking/cleaning/replacing the Power Carbon Brushes (565V DC)

Required time:	5 min.
Required tools	Standard tool kit; sliding caliper
Required materials	Carbon brush set power_F (3818254)

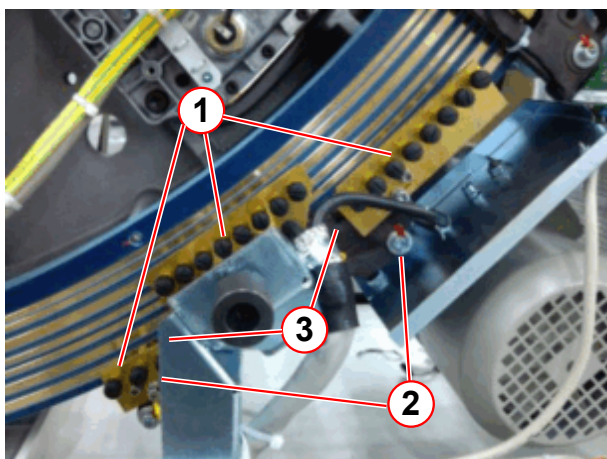
PM Power carbon brushes (565V DC) checked (6M)

PM Power carbon brushes (565V DC) replaced (6M)

Check the power carbon brushes (565V DC)

- Remove the screws caps and remove all power carbon brushes (565V DC).
- Measure each carbon brush with a sliding caliper.

Fig. 8: Power brush (565V DC) assembly



- (1) Power brushes (565V DC)
- (2) Two mounting screws
- (3) Centering holes

- Remove the two mounting screws and carefully bend away the assembly.
 - » The brush length must be greater than 17 mm to ensure the safe operation (14 mm limit) until the next maintenance interval.
- Check the connection points for oxidation/corrosion.

Cleaning / replacing the power carbon brushes (565V DC)

- Remove the carbon dust on the power brush assembly with a vacuum cleaner.
- Rotate the gantry manually until the center bolts can be inserted into holes in the slip ring.

Reinstall the power brush assembly by inserting the center bolts into the centering holes to position the brush assembly.

- » Insert the mounting screws with Loctite 243 applied and tighten the screws with **25 Nm**.
- Remove the centering bolts.

- Replace brushes if they are less than 17 mm or oxidation / corrosion is present.
- Insert the power carbon brushes.

2.10 Gantry: Checking/cleaning/replacing the Power Carbon Brushes (230V AC)

Required time:	5 min.
Required tools	Standard tool kit; sliding caliper
Required materials	Power brush (10415048) ¹

1. Ordering 3804692 can get both 10415047 and 10415048

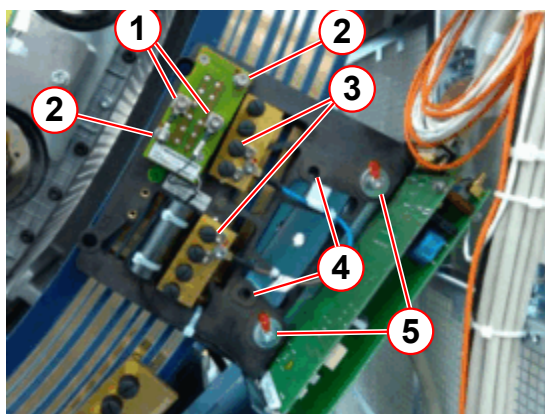
PM Power carbon brushes (230V AC) checked (6M)

PM Power carbon brushes (230V AC) replaced (6M)

Check the power carbon brushes (230V AC)

- Remove the screws caps and remove all power carbon brushes (230V AC).
- Measure each carbon brush(→ 3/ Fig. 9 Page 38) with a sliding caliper.
 - » The brush length must be greater than 16 mm to ensure the safe operation (12 mm limit) until the next maintenance interval.

Fig. 9: Power brush (230V AC) and signal brush assembly



- (1) X22 and X24
- (2) Two screws for the signal brush block
- (3) Power brushes (230V AC)
- (4) Centering holes
- (5) Two mounting screws for assembly

- Check the connection points for oxidation/corrosion.

Cleaning / replacing the power carbon brushes (230V AC)

- Remove the dust from the assembly using a vacuum cleaner.
- Replace brushes if they are less than 16 mm or oxidation / corrosion is present.
- Insert the power carbon brushes.

2.11 Gantry: Checking/cleaning/replacing the Signal Brush Block

Required time:	5 min.
Required tools	Standard tool kit; sliding caliper
Required materials	Signal brush block (10415047) ¹

1. Ordering 3804692 can get both 10415047 and 10415048.

PM Signal brush block checked (6M)

PM Signal brush block replaced (6M)

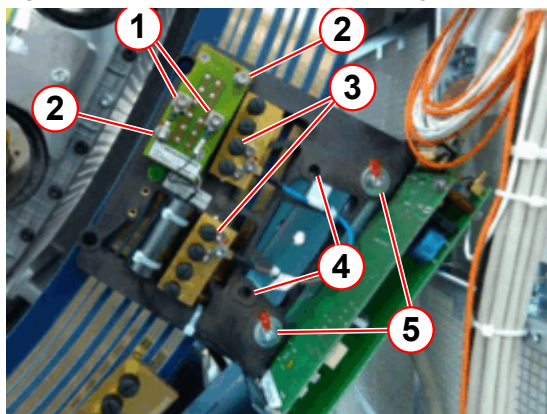
Check the signal brush block

- Disconnect X22 and X24 (→ 1/ Fig. 10 Page 39) from the signal brush block.
- Remove the two screws (→ 2/ Fig. 10 Page 39) to take out the signal brush block.
- Use a sliding caliper to measure from the end of the brush to the arrow-head (→ 1/ Fig. 11 Page 39).

The distance must be greater than 2 mm to ensure safe operation until the next maintenance interval.

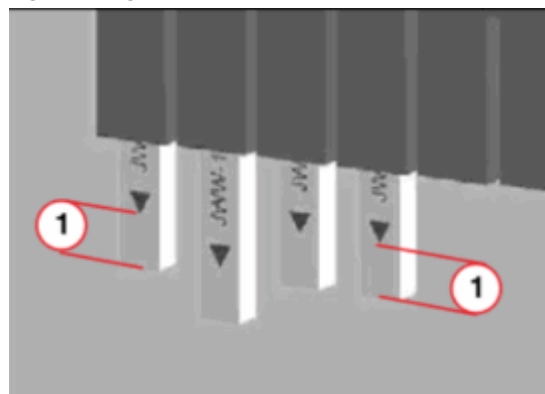
(The length of the carbon brush must not drop below the middle of the arrow before the next regular maintenance.)

Fig. 10: Power brush (230V AC) and signal brush assembly



- (1) X22 and X24
- (2) Two screws for the signal brush block
- (3) Power brushes (230V AC)
- (4) Centering holes
- (5) Two mounting screws for assembly

Fig. 11: Signal brush block



- (1) Distance to the arrowhead

- Check the connection points for oxidation/corrosion.

Cleaning / replacing the signal brush block

- Remove the dust on the signal brush block with a vacuum cleaner.
- Replace the block if the distance is less than 2 mm or oxidation / corrosion is present.

- Install the signal brush block and tighten it with **1.8 Nm**; install X22 and X24.

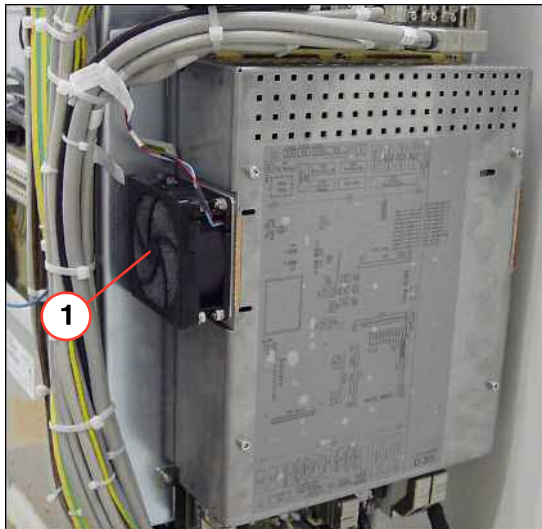
2.12 Gantry: Checking/replacing the MAS Air Filter

Required time:	5 min.
Required tools	Soft cloth
Required materials	Filter for MAS ventilator

PM Air filter checked (6M)

PM Air filter replaced (6M)

Fig. 12: MAS



(1) Filter cover for fan

- Remove the filter cover (pos.1) on the fan for the MAS.
- Check the filter for dust and replace or wash it, if necessary.
- Dry the filter with a cloth.
- Reinstall the filter.

2.13 IMS: Cleaning the ICS and the IRS

PM ICS and IRS cleaned (6M)

Required time:	15 min.
Required tools	Vacuum cleaner
Required materials	n.a.

- Clean the air inlets using a vacuum cleaner.

2.14 System: Installing the Covers



Read and observe the safety information contained in the "Safety Information" (→ Safety Information / Page 20) section of this document.

Required time:	20 min.
Required tools	Standard tool kit
Required materials	n.a.

SIM Covers and ground wire cables installed (6M)



All metal system covers have a plug or screw connection for the ground wire. Ensure that the ground wire cable is connected at this point while the system covers are installed. The ground wire cable is included with the components.

- Reinstall all covers.
- Switch on the system.

2.15 System: Cleaning the System

Required time:	5 min.
Required tools	Soft cloth; cleaning agent
Required materials	n.a.

SIM Flat screen cleaned (6M)

- Clean the flat screen.
- Remove any dirt or contamination caused during maintenance.

2.16 Switching on the System

Required time:	5 min.
Required tools	n.a.
Required materials	n.a.

2.17 System: Performing the Quality Check



Read and observe the safety information contained in the "Safety Information" section of this document(→ Safety Information / Page 20).



Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained.



Before executing the measurement, system check-up should have been performed.

Q Quality check completed (6M)

Required time:	25 min.
Required tools	Quality phantoms
Required materials	n.a.

The measurements are used for quality control and for checking the maintenance performed on the system.

The program routines may be selected via the **SERVICE/QUALITY** platform.

The **Constancy Test** automatically performs quality measurements.

- Click the **Constancy** function in the **Service/Quality** platform.

2.18 System: Checking System Functionality

Required time:	5 min.
Required tools	n.a.
Required materials	n.a.

Q Condition of the CT system checked (6M)

- Simulate several system functions typical for the customer to ensure that the system is in good working condition before turning it over to the customer.

2.19 Maintenance Documentation



You have completed this part of the maintenance. All work performed on the system during maintenance and the results obtained have to be documented in the "Maintenance Protocol" provided. Provide the customer with the complete protocol and, depending on organizational requirements, return a copy of it to your regional unit or field service.

2.20 End of Partial Maintenance

- » End of chapter

Maintenance type	Maintenance interval	Estimated total repair time:
Full maintenance	Every 12 months	300 min.



Read and observe the safety information contained in the "Safety information" (→ Safety Information / Page 20) section of this document.

Documents required

- Maintenance Protocol C2-015.832.01
- Replacement of parts (Gantry/PHS) C2-015.841.01

3.1 Switching on the System

Required time:	5 min.
Required tools:	n.a.
Required materials	n.a.

If the system is not in standby already, switch on the system.

- Switch on the LCB to power on the entire system.
- Switch on the PSB/UPS for the IMS.
- When the system reaches the standby mode, elevate the tabletop to its uppermost position and move it fully into the gantry.

3.2 Switching off the System

Required time:	2 min.
Required tools:	n.a.
Required materials	n.a.

Shut down the computers and power off the whole system and secure it against unintended switch on.

- In the ICS main user menu, select **System > End > Shutdown System** to shut down the computers.
- Switch off PSB/UPS and the power supply (on the LCB) to the entire system and secure it against unintended switch on.

3.3 Gantry: Cover Removal

Required time:	15 min.
Required tools:	Standard tool kit
Required materials	n.a.



Read and observe the safety information contained in the "Safety information" (→ Safety Information / Page 20) section of this document.



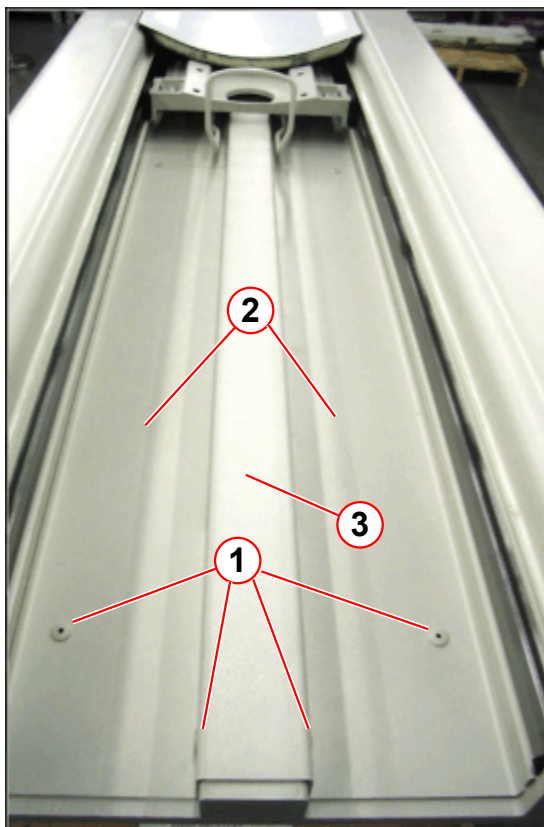
Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained.

- Release the tabletop and manually move it to its Home position.
- Remove the rot front and rot back assembly.
- Remove the filter assembly.
- Remove the right stand in back and the right stand of the main assembly.

3.4 PHS1A (P/N: 10355389): Cover Removal

Required time:	18 min.
Required tools:	Standard tool kit
Required materials	n.a.

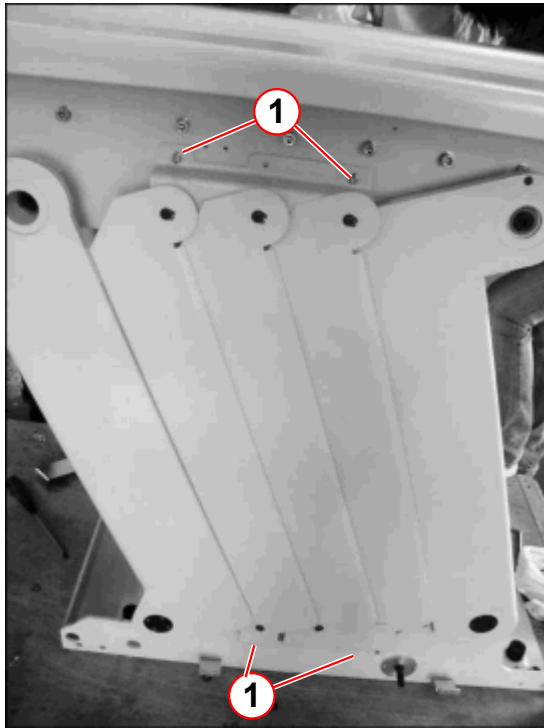
Fig. 13: Remove two back-end metal sheets



- (1) White screws
- (2) Metal sheets at the back-end
- (3) Metal bracket of the belt

- Remove two back metal sheets (pos.2) so that 6 lubricating nipples of the tabletop are accessible from underneath after the tabletop is moved to the back position.

Fig. 14: Cover plates



(1) 4 x screws

- Remove the back basin and center cover
- Remove the bottom cover.
- Remove the cover plate assembly with 4 screws (pos.1) so that the lubricating nipple of the vertical drive is accessible.

3.5 System: Visual Inspection according to DIN EN 62353/IEC 62353



Read and observe the safety information contained in the "Safety information" (→ Safety Information / Page 20) section of this document.

Required time:	10 min.
Required tools:	n.a.
Required materials	n.a.

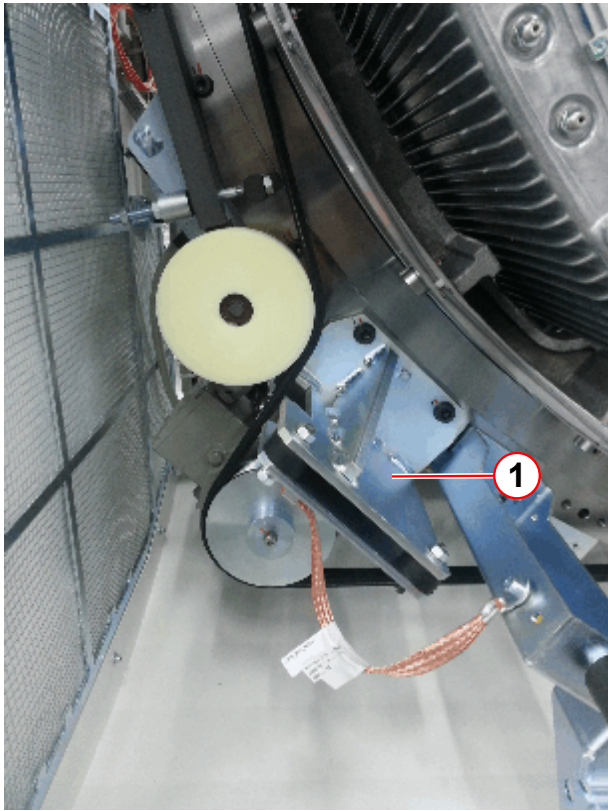
3.5.1 Gantry, Visual Inspection according to DIN EN 62353/IEC 62353

SI No safety-relevant damage (12M)

- Check the housing for safety-relevant damage:
 - All gantry housing components
 - Both gantry doors
 - The Makrolon strips in the gantry opening (scan plane)
 - » The cover panels must not have any visible deformation, sharp edges, or cracks.

- There are no joint cracks or aging cracks on the rubber damper of the motor mounting plate.

Fig. 15: Overview of motor mounting plate



(1) Motor mounting plate

Fig. 16: This is an example



3.5.2 PHS - Visual Inspection according to DIN EN 62353/IEC 62353

SI No safety-relevant damage (12M)

- Check the system housing for safety-relevant damage:
 - All patient table housing components
 - The upper section of the patient table, including the tabletop
 - » The cover panels must not have any visible deformation, sharp edges, or cracks.

3.5.3 System Cabinets, Visual Inspection according to DIN EN 62353/IEC 62353

SI No safety-relevant damage (12M)

- Check the system housing for safety-relevant damage:
 - All system cabinet housing components
 - » The cover panels must not have any visible deformation, sharp edges, or cracks.

3.5.4 Care Vision CT (optional), Visual Inspection according to DIN EN 62353/IEC 62353

SI No safety-relevant damage (12M)

- Check the system housing for safety-relevant damage:
 - All housing components of the support arm system (if applicable)
 - All housing components of the monitor trolley (if applicable)
 - » The cover panels must not have any visible deformation, sharp edges, or cracks.

3.5.5 System Cables and Cable Routing, Visual Inspection according to DIN EN 62353/IEC 62353



The check focuses exclusively on visible cables as well as their routing and ducts.

SI Cables and cable routing indicate no visible damage (12M)

- Go to the table. Check the connections identified by a dot (♦) for visible damage and to make sure the cable routing is correct (as applicable):
 - » The cable insulation must be intact and show no breaks.

Eliminate the possibility of persons tripping over cables. The cable must not be routed back and forth across the floor, or near sharp edges.

Cable connections between system components						
from ↔ to	LCB	PDS	Gantry	IRS	ICS/ Console	On-site supply
LCB	n.a.	♦	♦	n.a.	n.a.	♦
PDS	♦	n.a.	n.a.	n.a.	n.a.	n.a.
Gantry	♦	n.a.	n.a.	♦	♦	n.a.
IRS	n.a.	n.a.	♦	n.a.	♦	n.a.

Cable connections between system components						
ICS/ Console	n.a.	◆	◆	◆	n.a.	n.a.
On-site supply	◆	◆	n.a.	n.a.	n.a.	n.a.

3.5.6 Care Vision CT Cables and Cable Routing (optional), Visual Inspection according to DIN EN 62353/IEC 62353

SI Cables and ducts routed safely, no visible damage (12M)

- Check the cables for visible damage and make sure the routing is correct:
 - Cable to the support arm system (if applicable)
 - Cable to the monitor trolley (if applicable)
 - » The cable insulation must be intact and show no breaks.
- Eliminate the possibility of persons tripping over cables. The cable must not be routed back and forth across the floor, or near sharp edges.

3.5.7 Patient Positioning Aids, Visual Inspection according to DIN EN 62353/IEC 62353

SI In good shape, without visible damage (12M)

- Inspect the positioning accessories for damage (head rest, holder, pad).
- Check the accessory holder at the head end of the table for damage.
- Check the locking mechanisms of the patient positioning aids.
 - Attach the phantoms to the holder at the head end of the table, remove them, and repeat several times.
 - » The accessory must be attached securely to the patient table.



If the locking mechanism does not function properly, determine the cause of the problem (spring, knob, holder insert in the table, etc.), and replace the corresponding part(s).

3.5.8 Measurement Phantoms and Holder, Visual Inspection according to DIN EN 62353/IEC 62353

SI In good shape, without visible damage (12M)

- Inspect the measurement phantoms and holder for damage.
- Check the accessory holder at the head end of the table for damage.

- Check the locking mechanisms of the measurement phantoms.
 - Attach the phantoms to the holder at the head end of the table, remove them, and repeat several times.
 - » The accessory must be attached securely to the patient table.



If the locking mechanism does not function properly, determine the cause of the problem (spring, knob, holder insert in the table, etc.), and replace the corresponding part(s).

3.5.9 Osteo Accessory (optional), Visual Inspection according to DIN EN 62353/IEC 62353

SI In good shape, without visible damage (12M)

- Check the accessory for damage:
 - PE osteo pad
 - PE cushion insert
 - BMD cushion coupling
 - BMD reference phantom
 - » The accessory part must be in good shape and without visible damage.

3.5.10 Footswitch (optional), Visual Inspection according to DIN EN 62353/IEC 62353

SI In good shape, without visible damage (12M)

- Check the joystick for damage:
 - » The footswitch must be in good shape and without visible damage.

3.5.11 Joystick (optional) - Visual Inspection according to DIN EN 62353/IEC 62353

SI In good shape, without visible damage (12M)

- Check the joystick for damage:
 - » The joystick must be in good shape and without visible damage.

3.6 Mobile Kit (option): Inspections

Required time:	60 min.
Required tools:	Torque wrench 25-130 Nm
Required materials	n.a.

The mobile CT requires additional maintenance due to extraordinary conditions.

3.6.1 Transportation Lock Inspection

SI Inspection for damages (12M)

- Visually inspect for damages and test their functions.

3.6.2 Table Inspection

SI Inspect for damages and loose parts (12M)

- Visually inspect for damage, loose screws and parts as well as cracks/tears, especially at stressed parts.
- To check attachment of the PHS to adequate flooring, please refer to the guidelines of the trailer manufacturer (refer to reference in the system-specific Planning Guide C2-015.891.01.xx.02).

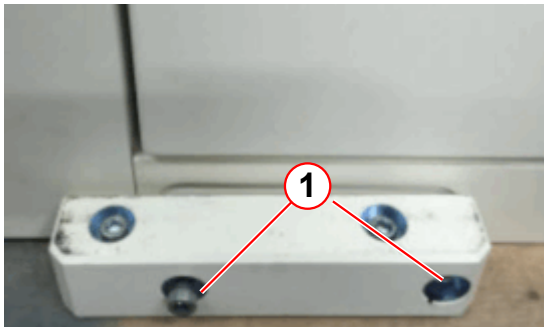
3.6.3 Gantry Inspection

SI Inspect for damages, loose parts, cracks/tears, screw connections (12M)

- Visually inspect for:
 - Damages
 - Loose screws and parts
 - Cracks / tears especially at stressed parts and along visible welding seams, particularly at the gantry base

- Inspect the following using a torque wrench:

Fig. 17: Block at gantry stand



(1) Screws

The screw connection between each one of the 4 blocks and the gantry (torque 90 Nm)

Fig. 18: Pedestal

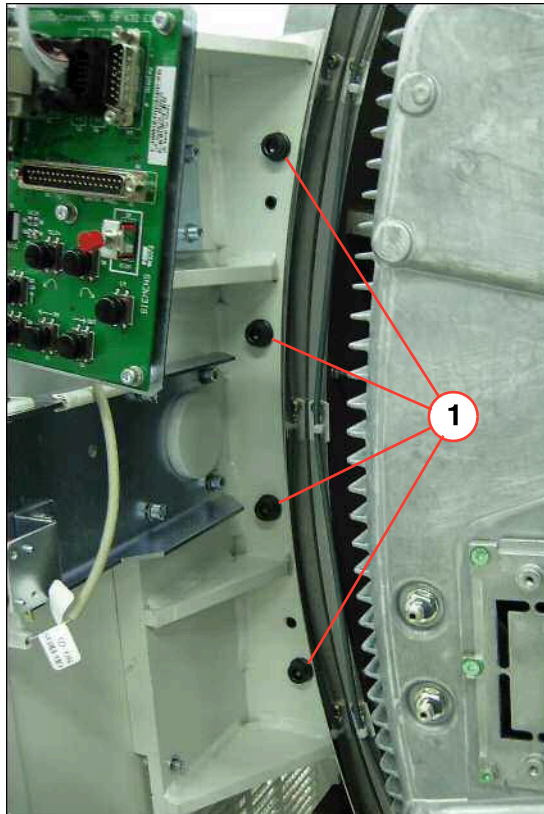


(1) Screws

The left and right-hand side pedestal (torque 50 Nm) - 2 screws (pos.1) on each side

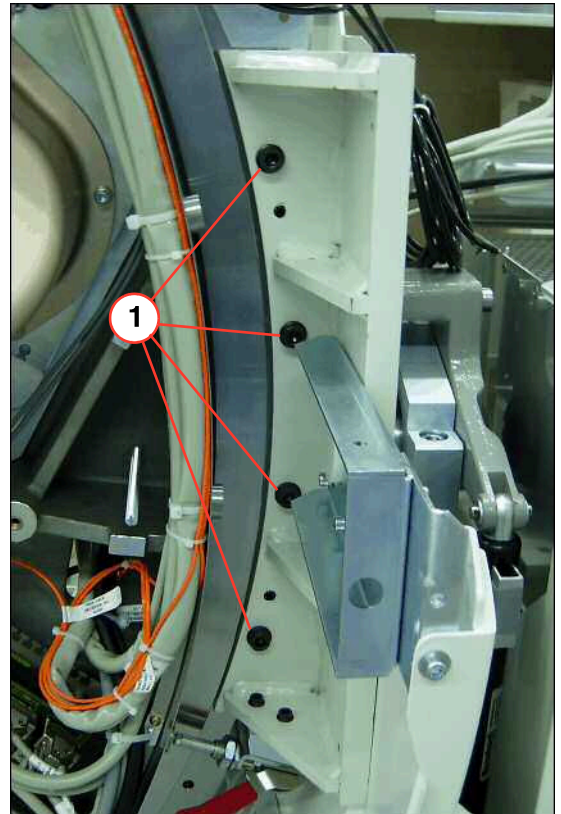
- The left and right hand side tilt suspension (torque 90 Nm) - 4 screws (pos.1) on each side

Fig. 19: Tilt suspension (left hand side)



(1) Screws

Fig. 20: Tilt suspension (right hand side)



(1) Screws

- The X-ray tube (torque 90 Nm)
- The high voltage generator (torque 90 Nm)
- To check the attachment of the gantry to adequate flooring, please refer to the guidelines of the trailer manufacturer (refer to the references in the system-specific Planning Guide C2-015.891.01.xx.02).
- If necessary, vacuum clean the cover to remove synthetic particles produced by the cover rubbing against the collimator during transportation.

3.7 System: Label Inspection

Required time:	2 min.
Required tools:	n.a.
Required materials	n.a.

SI Labels inspected (12M)

If any label has peeled off the system or a component, replace the label. To identify the label as well as the correct location of it, refer to the **Location of Labels** section in the **System Owner Manual** which has to be stored at all times with the system.

3.8 PHS1A (P/N:10355389): Lubricating the vertical Drive Spindle

PM Vertical drive spindle lubricated (12M)

Required time:	2 min.
Required tools:	Soft cloth; grease gun; Tungaloy B.S.; Topas NCA 52
Required materials	n.a.



Since March 2015, a new vertical drive motor with 2 grease fittings has been introduced in PHS 1A.

If the vertical drive motor only contains 1 grease fitting, please follow the instructions below.

Tab. 3 Vertical drive motor with 1 grease fitting

Fig. 21: Vertical drive spindle

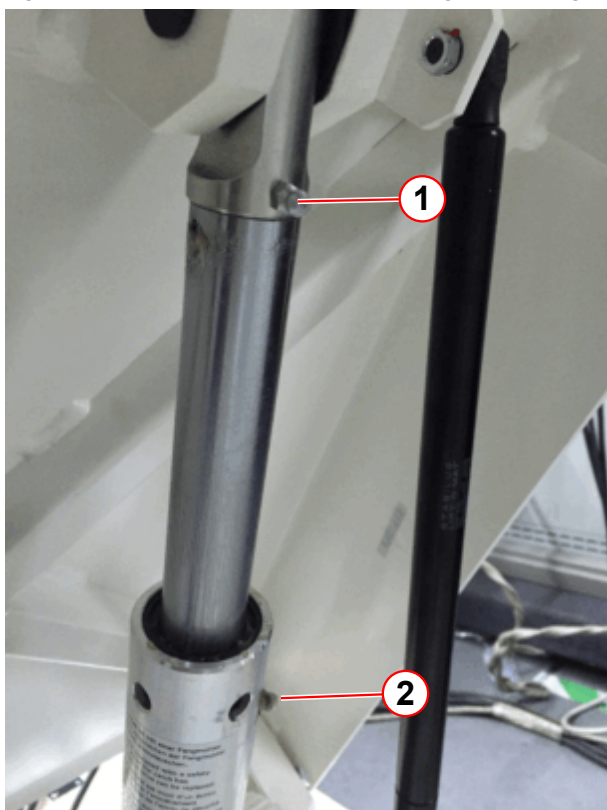


(1) Lubricating nipple

- Place the connecting piece of the grease gun containing Tungaloy B.S. on the grease fitting (1)
- Insert approximately **15 g** of lubricant.

If the vertical drive motor contains 2 lubricating nipples, please follow the instructions below.

Fig. 22: New vertical drive motor with 2 grease fittings



1. Place the connecting piece of the grease gun containing Tugrease B.S. on the lubricating nipple **(1)**. Insert approximately **15 g** of lubricant.
2. Place the connecting piece of the grease gun containing Topas NCA 52 on the lubricating nipple **(2)**. Insert approximately **2 g** of lubricant.

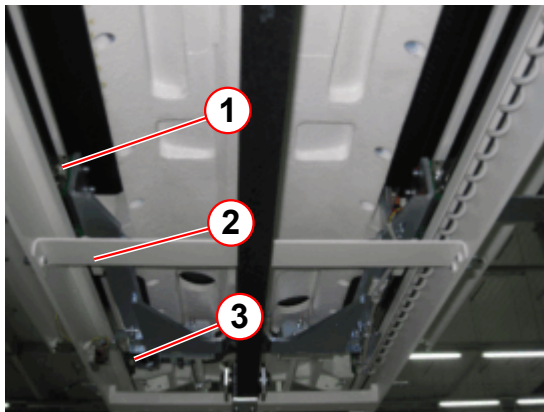
- (1) Top grease fitting
(2) Bottom grease fitting

3.9 PHS1A (P/N:10355389): Lubricating the Guide Rails of the Table

PM Guide rails lubricated (12M)

Required time:	3 min.
Required tools:	Soft cloth; grease gun; Isoflex Topas NCA 52
Required materials	n.a.

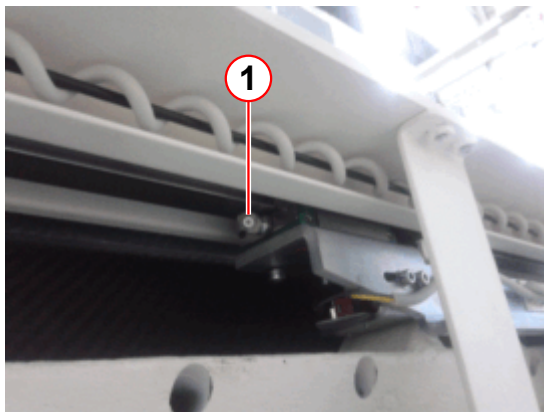
Fig. 23: Access the nipples from underneath (picture bottom is PHS1A rear end)



- (1) The nipple on one side (another symmetric one is on the side)
- (2) The nipple not shown in the picture (another symmetric one is on the other side)
- (3) The nipple on one side (another symmetric one is on the other side)

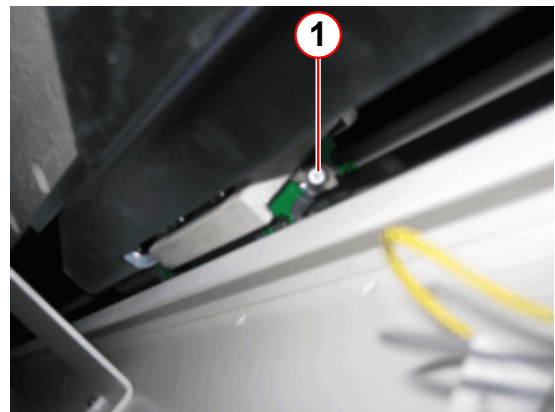
- Move the table to the back position so that the 6 lubricating nipples are accessible from underneath.
- Place the connection piece of the grease gun with lubricant Isoflex Topas NCA 52 on the lubricating nipple (2) from underneath.
- Press the lubricant into the carriage until new lubricant can be seen on the guide rail.
- Clean old lubricant off the guide rails with a soft cloth .
- Press a little lubricant on a soft clean cloth.
- Use this cloth with new lubricant to clean the guide rails.

Fig. 24: Local view of the nipple



- (1) One of the nipples

Fig. 25: Local view of the nipple



- (1) One of the nipples

3.10 PHS1A (P/N:10355389): Checking the Horizontal Belt Tension

PM Belt tension check (12M)

Required time:	15 min.
Required tools:	Standard tool kit; ruler; spring scale
Required materials	n.a.

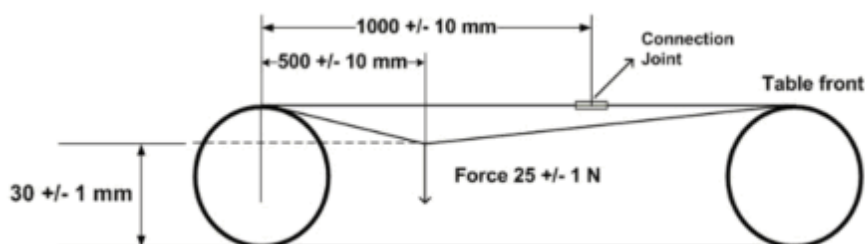


The drive belt tension should be checked only once after the first PHS installation, replacing the tabletop drive belt, horizontal drive and toothed belt.

Check the horizontal drive belt tension:

- Move the table forward 1000 +/- 10 mm so that the distance between the tension adjuster and the connection joint is 1000 +/- 10 mm.
- Loosely install a cable tie around the belt and position it approx. in the center of the tension adjuster and the connection joint ($1000\text{mm}/2 = 500\text{mm}$).
- Hook a spring scale (100 N) into the cable tie.
- Position a ruler or measuring tape in the vertical position next to the cable tie.
- Pull the spring scale in the vertical direction with a tension of 25 +/- 1 N and measure the vertical change in distance (see below illustration).
 - » The vertical change in deflection must be 30, +/-1 mm (at 25, +/- 1 N).
- Adjust the belt tension accordingly with the adjustment screw until it is within the tolerance margin.

Fig. 26: Belt tension



3.11 PHS1A (P/N:10355389): Checking the anchoring of the PHS

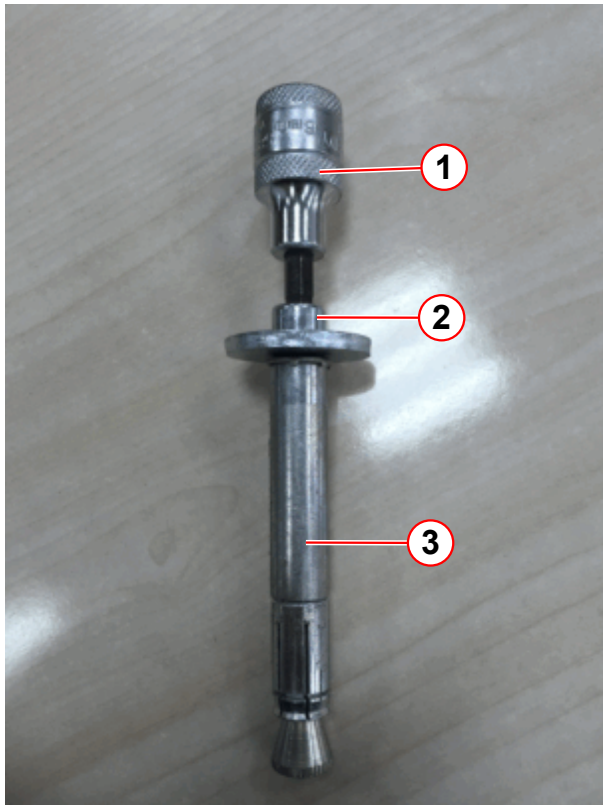
PM Anchoring check (12M)

Required time:	10 min.
Required tools:	Torque wrench 25 - 130 Nm
Required materials	n.a.

Check the torque of the 4 floor mounting screws:

- The LIEBIG anchors (M8) must be secured with a torque of 25 Nm.
Use a torque wrench with a size 6 mm hex bit socket.

Fig. 27:



- (1) 6 mm hex bit socket
- (2) Screw M8 X 25/35 (Strength: 8.8)
- (3) Liebig self-undercut anchor IL S 14/80 with a 14 mm drill bit

- The HILTI anchors (M10) must be secured with a torque of 50 Nm.
Use a torque wrench with a size 17 mm socket.

Fig. 28:



(1) 17 mm socket

(2) Hilti HSL- 3-G M10/100 with a 15 mm drill bit

- If you cannot obtain the recommended torque, check the floor condition according to the "Planning Guide" and the "Installation and Setting Instructions." Contact the corresponding project manager for on-site information.

3.12 MCS (option): Checking the Ceiling System

This chapter only applies for Ondal Systems, MCS 1, Mat. No. 10430762 and MCS 2, Mat. No. 10430763 (delivery started in 2010)

Required time:	15 minutes without adjustments
Required tools:	n.a.
Required materials	n.a.

3.12.1 Visual Checks of the Monitor Ceiling System

SI Visual checks for varnish damage (12M)

- Check the MCS for paint damage.

SI Visual checks for cracks (12M)

- Check the MCS for cracks

3.12.2 Functional Checks of the Monitor Ceiling System

PM Horizontal and vertical movements checked (12M)

- Move the spring arms in the horizontal and vertical direction. Smooth movement is required in all directions.

PM Mechanical end stops checked (12M)

- Move the MCS to the mechanical end stops. The end stops have to block the movement.

PM Load balance checked (12M)

- Move the MCS to several positions and then release it. The support arm must keep its position over the entire lift range.

In case the support arm is moving after its release, the brake has to be adjusted according to the document "Installation of Options".

3.13 Sliding gantry (option): Checking

Required time:	15 min.
Required tools:	Standard tool kit; torque wrench, 1-12 Nm
Required materials	n.a.

PM Check toothed belt (12M)

- The toothed belt must be **checked annually**.
 - » To adjust the tension of the toothed belt, tighten each tensioning screw with a torque of **1.8 +0.2 Nm for a belt route length of ≤ 3.2 m**.



For adjustment, the gantry must be positioned fully toward the side of the drive motor.



Make sure that the adjustment screws are easy to loosen (add oil or grease if necessary).

Fig. 29:

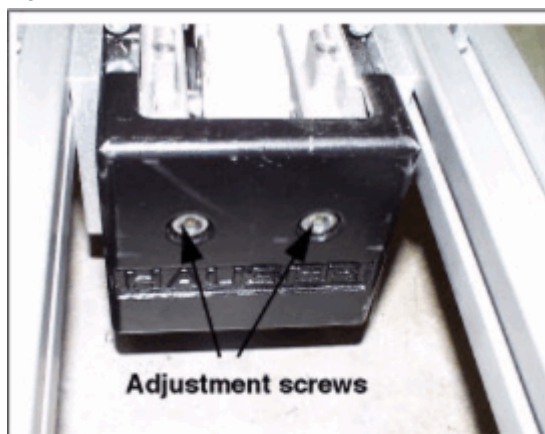


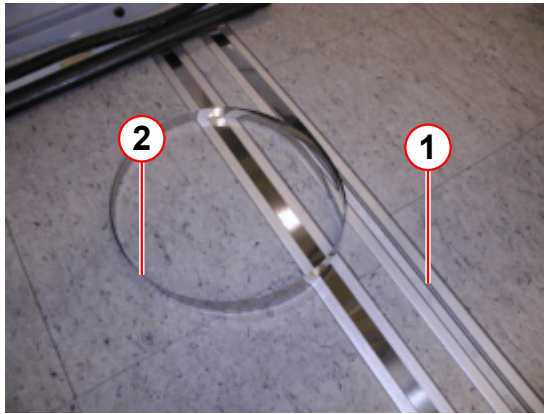
Fig. 30:



The toothed belt must run centered on the toothed wheel. Adjust it in small steps alternately with both tensioning screws. Move the gantry in both directions and check to make sure the belt runs centered on the wheel.

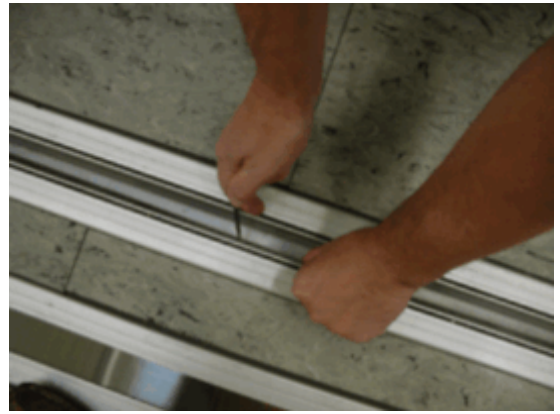
- Check to make sure there are no foreign particles on the rails.
 - » Open the metal covering as shown in the image and check the rail inside.

Fig. 31: Rails



- (1) Rail; illustration shows two rails for Sensation; for Emotion, there is only one rail
 (2) Metal covering

Fig. 32: Checking the rail inside



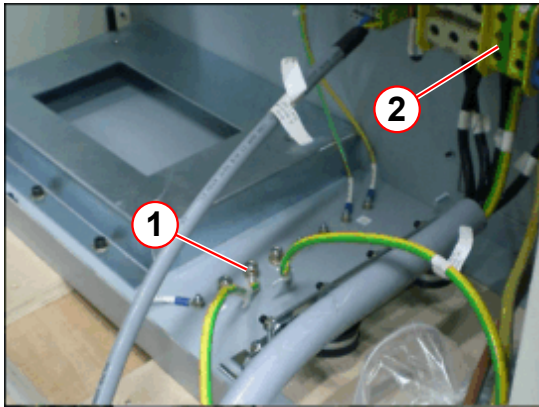
PM Check switching elements, front and rear (12M)

- Check the switching elements by moving the gantry in the forward and reverse directions.
 - » Push your foot against the switching element to check the stop function.

PM Check protective conductor connection (12M)

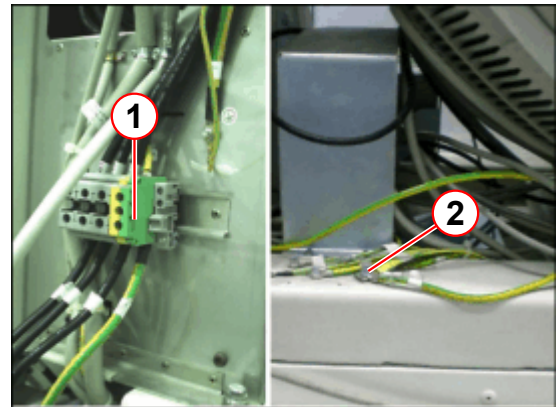
- Check both protective conductor cables between the line connection box (LCB) and the right stand of the gantry.
 - » Disconnect both protective conductor cables (in W800 and W801 PE) in the LCB for the measurement (→ Fig. 33 Page 74) (to check the protective conductor cable for possible cable breaks in the energy chain).
 - » Each protective conductor cable must be measured separately between the LCB and the gantry (→ Fig. 34 Page 74).
 - » Reconnect the protective conductor cables.

Fig. 33: Protective conductor cables, LCB



- (1) Cable W801, in LCB
- (2) Productive conductor cable of W800, in LCB

Fig. 34: Protective conductor cables, gantry



- (1) Protective conductor terminal of W800, PDS front
- (2) Protective conductor terminal of W801, gantry rear

3.14 Gantry: Checking the Drive Belt Tension

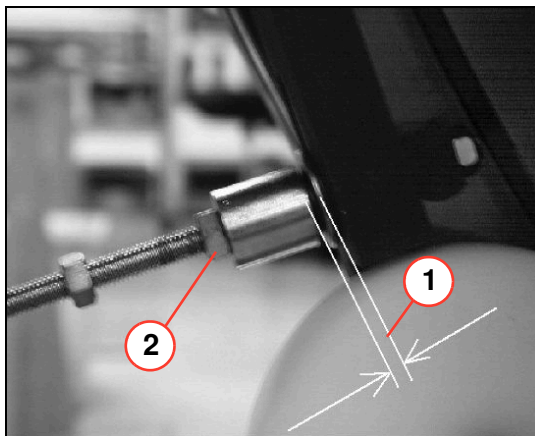


The drive belt tension has to be measured during system installation, each time maintenance is performed, and after replacement of the drive belt. Additional measurements are unnecessary.

PM Drive belt tension check (12M)

Required time:	5 min.
Required tools:	Standard tool kit; Feeler gauge;
Required materials	n.a.

Fig. 35: Drive belt tension



- (1) Distance
(2) Adjustment screw

- Measure the distance (1) between the washer and the edge of the cylinder with a feeler gauge as shown.
 - » The distance has to be between **2.8 mm ± 0.1**.
- If necessary, adjust the distance with the adjustment screw (2).

3.15 Gantry: Checking the Gantry Brake



A loose gantry brake may lead to safety-related risks, e.g., during tube replacement. The gantry brake has to be checked during system installation and each time maintenance is performed on the system.

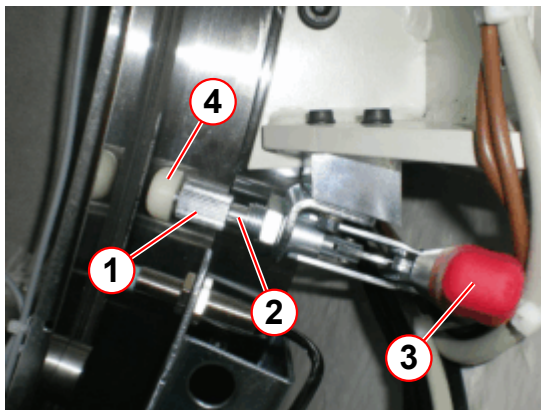
PM Gantry brake checking (12M)

Required time:	5 min.
Required tools:	n.a.
Required materials	n.a.

Follow the steps below to check the gantry brake:

- Switch off the system and secure it against unintended switch on.
- Lock the gantry brake.
- Using reasonable force, carefully push the HV tank clockwise to check if the gantry disk rotates.
 - » If the disk rotates, the gantry brake has to be adjusted.

Fig. 36: Gantry brake



- (1) Lock nut of brake shaft
- (2) Brake shaft
- (3) Brake handle
- (4) Brake cap

If necessary, adjust the gantry brake as shown in the steps below:

- Unlock the brake (3) and loosen the brake shaft lock nut (1) until the brake cap (4) screw is moveable by hand.

Note: the shaft lock nut (1) may look different depending on how the system was assembled.
- Turn the brake cap (4) away from the gantry belt to leave a small gap between the brake cap and the gantry belt.
- Lock the brake (3), and screw in the brake cap (4) until the brake cap (4) touches the gantry belt.
- Unlock the brake, and turn the brake cap (4) clockwise toward the gantry belt one more time to assure tight attachment.
- Tighten the brake shaft lock nut (1).
- Lock the brake to test the results of the adjustment.

3.16 Gantry: Checking the Cover Hooks

PM Check the elasticity of the hook holders on both the front and rear of the gantry (12M)

Required time:	1 min.
Required tools:	n.a.
Required materials	n.a.

- Push the hooks into the holders.
- Pull out the hooks and push them in again.

3.17 Gantry: Cleaning the Air Outlets on the Gantry Top

PM Air outlets cleaned (12M)

Required time:	5 min.
Required tools:	Vacuum cleaner
Required materials	n.a.

- Remove the dust from the grid using a vacuum cleaner.

3.18 Gantry: Checking the Gantry Cooling Fans

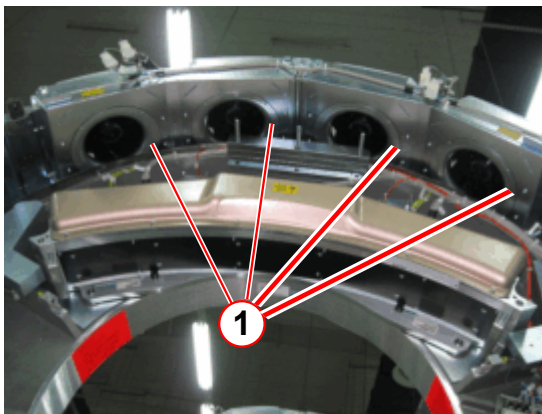
PM Gantry cooling fans checked (12M)

PM Gantry cooling fans checked (12M)

Required time:	2 min.
Required tools:	n.a.
Required materials	n.a.

- Use your hand to rotate the fans' blades to see if rotation is smooth and if there is noise.
- Check manually if the rotational part shifts on the axle.
- Compare the rotation speed of the fans to see if they are similar to each other.
- If there is any deviation from above inspections, replace the corresponding fan or fans.

Fig. 37: Gantry cooling fans



(1) Fans

3.19 Gantry: Cleaning the Slip Rings

PM Slip ring assembly cleaned (12M)

Required time:	5 min.
Required tools:	Vacuum cleaner
Required materials	n.a.

- Remove the carbon dust from the slip ring using a vacuum cleaner while rotating the gantry step-by-step counter-clockwise by hand.



Do not touch the surface of the slip ring with your fingers or with parts of the vacuum cleaner.

3.20 Gantry: Checking/cleaning/replacing the Power Carbon Brushes (565V DC)

Required time:	5 min.
Required tools	Standard tool kit; sliding caliper
Required materials	Carbon brush set power_F (3818254)

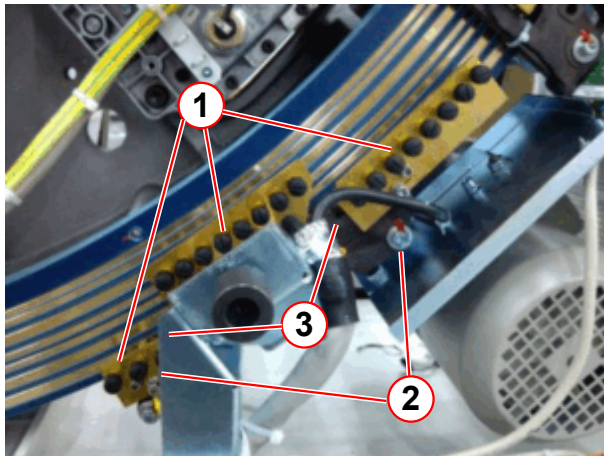
PM Power carbon brushes (565V DC) checked (6M)

PM Power carbon brushes (565V DC) replaced (6M)

Check the power carbon brushes (565V DC)

- Remove the screws caps and remove all power carbon brushes (565V DC).
- Measure each carbon brush with a sliding caliper.

Fig. 38: Power brush (565V DC) assembly



- (1) Power brushes (565V DC)
- (2) Two mounting screws
- (3) Centering holes

- Remove the two mounting screws and carefully bend away the assembly.
 - » The brush length must be greater than 17 mm to ensure the safe operation (14 mm limit) until the next maintenance interval.
- Check the connection points for oxidation/corrosion.

Cleaning / replacing the power carbon brushes (565V DC)

- Remove the carbon dust on the power brush assembly with a vacuum cleaner.
- Rotate the gantry manually until the center bolts can be inserted into holes in the slip ring.

Reinstall the power brush assembly by inserting the center bolts into the centering holes to position the brush assembly.

- » Insert the mounting screws with Loctite 243 applied and tighten the screws with **25 Nm**.
- Remove the centering bolts.

- Replace brushes if they are less than 17 mm or oxidation / corrosion is present.
- Insert the power carbon brushes.

3.21 Gantry: Checking/cleaning/replacing the Power Carbon Brushes (230V AC)

Required time:	5 min.
Required tools	Standard tool kit; sliding caliper
Required materials	Power brush (10415048) ¹

1. Ordering 3804692 can get both 10415047 and 10415048

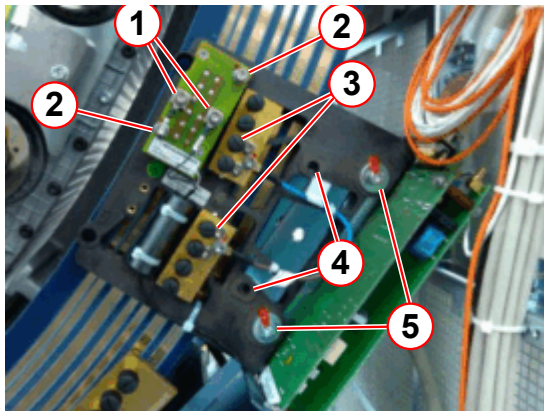
PM Power carbon brushes (230V AC) checked (6M)

PM Power carbon brushes (230V AC) replaced (6M)

Check the power carbon brushes (230V AC)

- Remove the screws caps and remove all power carbon brushes (230V AC).
- Measure each carbon brush(→ 3/ Fig. 39 Page 83) with a sliding caliper.
 - » The brush length must be greater than 16 mm to ensure the safe operation (12 mm limit) until the next maintenance interval.

Fig. 39: Power brush (230V AC) and signal brush assembly



- (1) X22 and X24
- (2) Two screws for the signal brush block
- (3) Power brushes (230V AC)
- (4) Centering holes
- (5) Two mounting screws for assembly

- Check the connection points for oxidation/corrosion.

Cleaning / replacing the power carbon brushes (230V AC)

- Remove the dust from the assembly using a vacuum cleaner.
- Replace brushes if they are less than 16 mm or oxidation / corrosion is present.
- Insert the power carbon brushes.

3.22 Gantry: Checking/cleaning/replacing the Signal Brush Block

Required time:	5 min.
Required tools	Standard tool kit; sliding caliper
Required materials	Signal brush block (10415047) ¹

1. Ordering 3804692 can get both 10415047 and 10415048.

PM Signal brush block checked (6M)

PM Signal brush block replaced (6M)

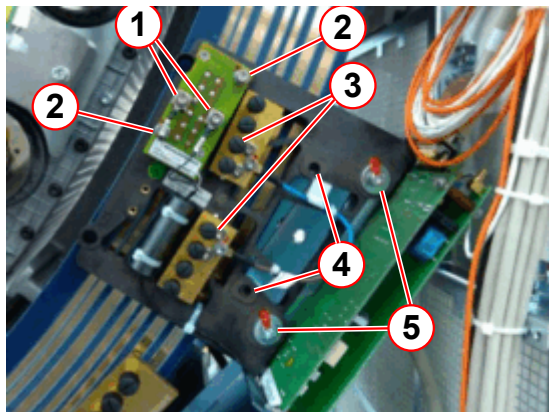
Check the signal brush block

- Disconnect X22 and X24 (→ 1/Fig. 40 Page 84) from the signal brush block.
- Remove the two screws (→ 2/Fig. 40 Page 84) to take out the signal brush block.
- Use a sliding caliper to measure from the end of the brush to the arrow-head (→ 1/Fig. 41 Page 84).

The distance must be greater than 2 mm to ensure safe operation until the next maintenance interval.

(The length of the carbon brush must not drop below the middle of the arrow before the next regular maintenance.)

Fig. 40: Power brush (230V AC) and signal brush assembly



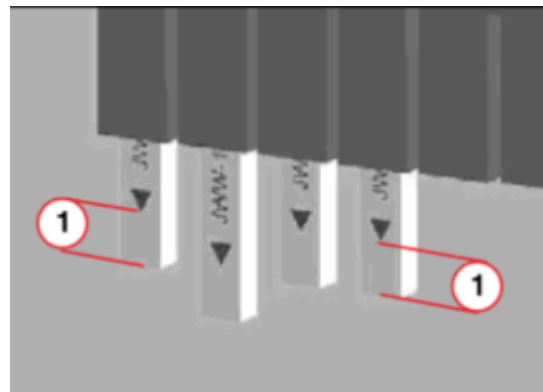
- (1) X22 and X24
- (2) Two screws for the signal brush block
- (3) Power brushes (230V AC)
- (4) Centering holes
- (5) Two mounting screws for assembly

- Check the connection points for oxidation/corrosion.

Cleaning / replacing the signal brush block

- Remove the dust on the signal brush block with a vacuum cleaner.
- Replace the block if the distance is less than 2 mm or oxidation / corrosion is present.

Fig. 41: Signal brush block



- (1) Distance to the arrowhead

- Install the signal brush block and tighten it with **1.8 Nm**; install X22 and X24.

3.23 Gantry: Checking/replacing the Air Filters for the Gantry

Required time:	15 min.
Required tools:	Standard tool kit
Required materials	Gantry filters

PM Air filter checked (12M)

PM Air filter replaced (12M)

Fig. 42: Filter cover



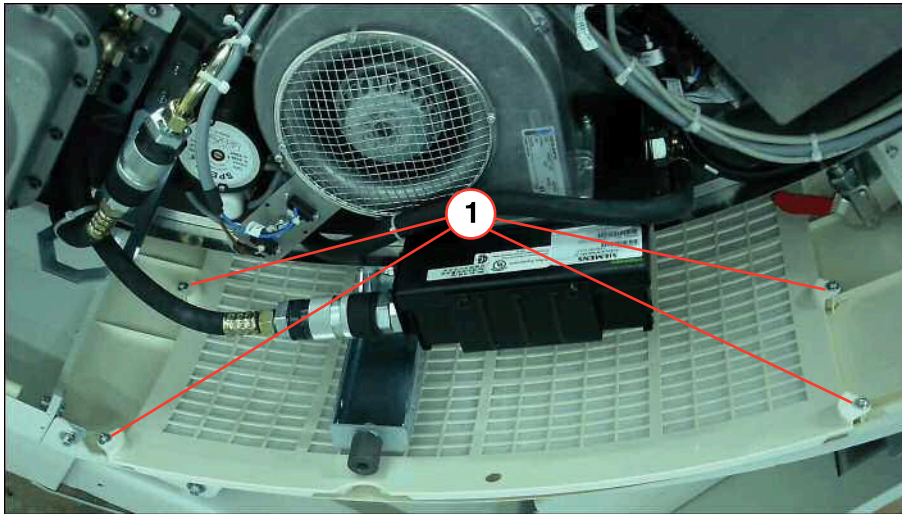
- Check the filter at the back of the gantry and replace it, if necessary:

- Remove the gantry back cover and the filter assembly. (refer to "Replacement of parts Gantry/PHS", print no. C2-015.841.01, chapter "General")
- Loosen the screws on the filter assembly, and replace the filter.
- Close the covers.

- Check the filter at the right bottom of the gantry and replace it, if necessary:
 - Remove the gantry front cover (refer to "Replacements of Parts, Gantry/PHS", print no C2-015.841.01, chapter "General").

- Remove the screws on the filter assembly.

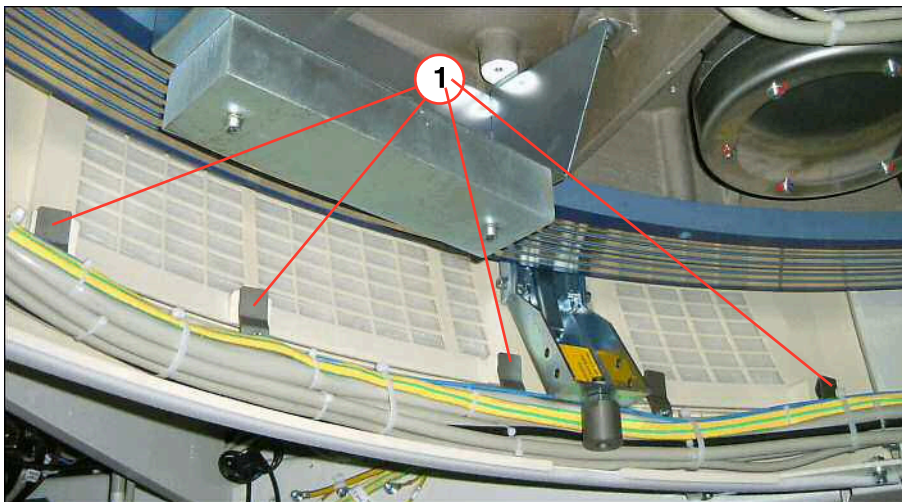
Fig. 43: Right lower filter in the gantry (front view)



(1) Screws that need to be removed

- Pull the filter assembly towards the gantry front until it is out of the gantry.
- Loosen the rubber band at the back of the filter assembly and remove the filter.
- Fix the new filter to the filter cover with the rubber band, and install the filter assembly to the gantry. Make sure that the filter is positioned under the clips at the back side of the gantry.

Fig. 44: Right lower filter (rear view)

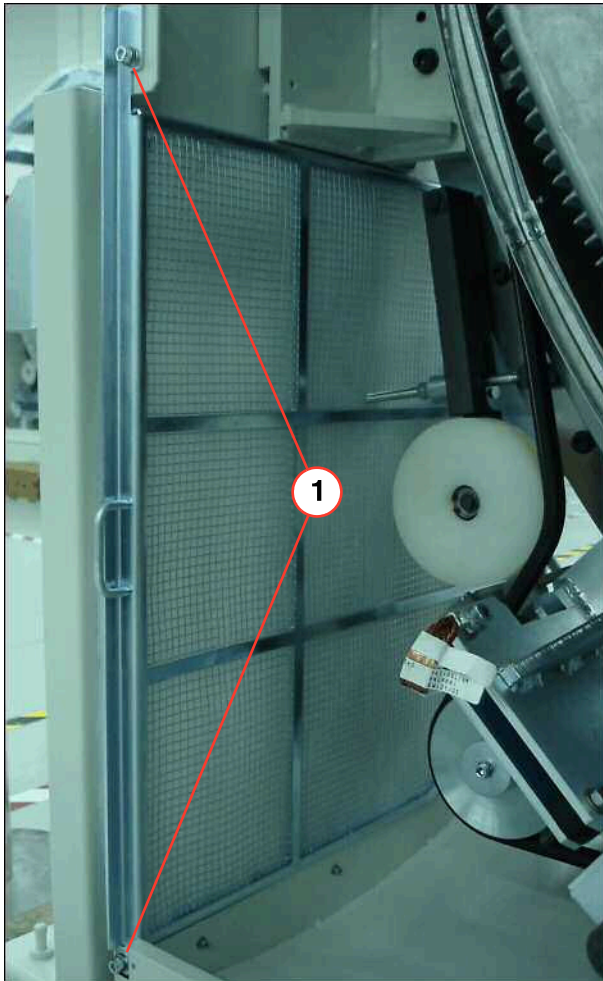


(1) Clips on the filter

- Close the gantry front cover.

- Check the filter at the left side of the gantry and replace it, if necessary:

Fig. 45: Left filter in the gantry



(1) Screws that need to be removed

- Remove the gantry front cover (refer to "Replacements of Parts, Gantry/PHS", print no C2-015.841.01, chapter "General").
- Replace the filter and attach it to the gantry.
- Close the covers.

3.24 Gantry: Lubricating the Main Bearing

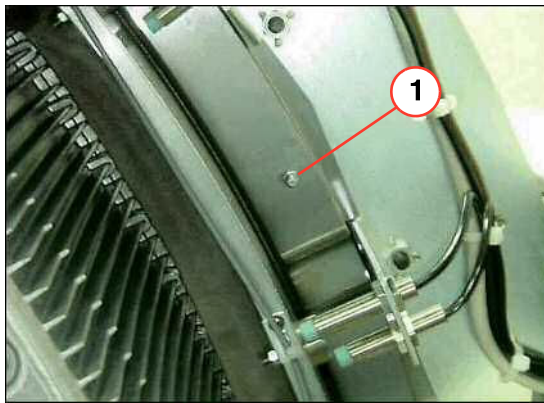
Required time:	10 min.
Required tools:	Standard tool kit; grease gun; Isoflex Topas NCA 52 lubricant
Required materials	n.a.

PM Main bearing lubricated (12M)



Approximately 70~80g of the lubricant Isoflex Topas NCA 52 should be applied. Determine in advance how much lubricant is released each time the lubricant gun is applied. In this way you will know how much lubricant to apply.

Fig. 46: Lubricating nipple



- The lubricating nipple is located next to the rotation sensors.
 - Connect the grease gun to lubricating nipple (pos.1) directly.
-
- Apply approximately 12g of lubricant.
 - Rotate the gantry clockwise by hand (approximately 120°).
 - Apply another (approx.) 12g of lubricant.
 - Rotate the gantry by hand again (approximately 120°) in clockwise direction.
 - Apply another (approx.) 12g of lubricant.
 - Rotate the gantry several turns by hand in the clockwise direction.
 - Repeat the above procedures
 - Disconnect the grease gun from the lubricating nipple.
 - Check for and remove excess lubricant at the bearing.
 - Rotate the gantry in a slow mode when starting the system.

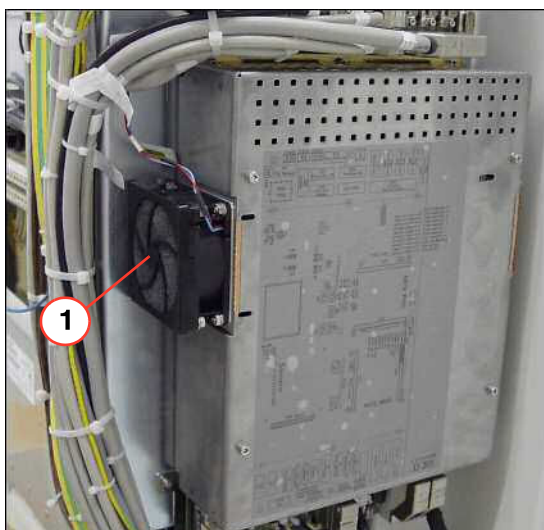
3.25 Gantry: Checking/replacing the MAS Air Filter

Required time:	5 min.
Required tools:	n.a.
Required materials	MAS air filter

PM Air filter checked (12M)

PM Air filter replaced (12M)

Fig. 47: MAS



(1) Filter cover for fan

- Remove the filter cover (pos.1) on the fan for the MAS.
- Check the filter for dust and replace or wash it, if necessary.
- Dry the filter with a cloth.
- Reinstall the filter.

3.26 Gantry: Cleaning the Tube Cooler

Required time:	10 min.
Required tools:	Standard tool kit; vacuum cleaner
Required materials	n.a.

PM Tube cooler cleaned (12M)

Fig. 48: Tube cooling system



- Rotate the gantry so that the tube cooler is in a comfortable working position.
- Secure the gantry with the mechanical brake.
- Remove the dust from the tube cooler using a vacuum cleaner.

3.27 IMS: PSB/UPS Checking and ICS/IRS Cleaning

Required time:	15 min.
Required tools:	Vacuum cleaner
Required materials	n.a.

PM PSB/UPS position checked (12M)

- Switch off the system and secure it against unintended switch on.
- Make sure that the PSB/UPS is **NOT** on top of the ICS and IRS.



Strictly follow the conditions described below:

- Do NOT place the PSB/UPS on top of the ICS and IRS. Otherwise, it will damage the computers because of heat accumulation.
- The PSB/UPS will be positioned between the ICS and IRS (leave a space between them) in the IMS cabinet.

PM ICS and IRS cleaned (12M)

- Clean the air inlets using a vacuum cleaner.

3.28 System: Measuring the Protective Conductor Resistance

Required time:	20 min.
Required tools:	Protective conductor meter
Required materials	n.a.

Note: All of the following "to" points should be accessible, protective grounded metal parts of the CT component.

- Use a protective conductor meter to perform the test and enter the data of that measuring device at the beginning of the protocol.
- To ensure correct results, unroll the test cable completely from the testing device prior to performing the test.
- **Enter the value measured into the corresponding protocol or checklist.**

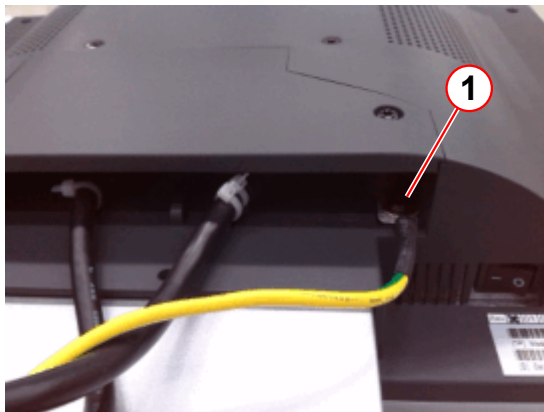
3.28.1 General Notes when performing Measurements at the Monitors

Always follow the warning in the note shown below when measuring at one of the monitors.



Disconnect the monitor signal cable from the computer. Otherwise, the graphics board will be destroyed.

Fig. 49: 19 inch flat screen - 10520199



(1) Grounding terminal; rear

3.28.2 SOMATOM Scope

SIE LCB: protective conductor resistance $\leq 100 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: uncoated metallic part

SIE Gantry: protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Uncoated metallic part

SIE Patient table: protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Uncoated metallic part

SIE ICS Tower PC: protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Uncoated metallic part

SIE ICS monitor: protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Measurement point of the respective monitor

SIE PSB/UPS for ICS: protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: PSB/UPS protective ground

SIE IRS Tower PC: protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Uncoated metallic part

SIE Ceiling support (in scan room, option): protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Uncoated metallic part

SIE Monitor trolley (in scan room, option): protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Uncoated metallic part

SIE Monitor (in scan room, option): protective conductor resistance $\leq 300 \text{ m}\Omega$ (12M)

- From: protective conductor terminal in the on-site power distribution
To: Measurement point of the respective monitor

3.29 Switching on the System

Required time:	5 min.
Required tools:	Standard tool kit
Required materials	n.a.

- Switch on the LCB to power on the entire system.
- Switch on the PSB/UPS for the IMS.

3.30 System: Function Tests

Required time:	30 min.
Required tools:	n.a.
Required materials	n.a.

To perform the following function tests, the system has to be switched on and in working condition.

3.30.1 Checking the Emergency STOP Circuit of the System

SI Emergency STOP circuit checked (12M)



All motor-driven system movements must shut down when the Emergency STOP button is pressed.

- Enable the service functions by selecting: **Options > Local Service > Test Tools**.
- Select a **Rot Mode**.
- Select: **GO**
- The Rot mode is loaded and gantry rotation starts automatically.
- Push the red **STOP** button at the control box.
- Gantry rotation has to end and come to a complete stop.
- The displays on the gantry control panels have to be off.
- The NOTE window is displayed on screen
 - Select: **Continue** (in the NOTE window)
- Follow the instructions in the message window of Rot mode.
- The system has to automatically return to Standby.
 - Wait until the message: **"RotStatic is idle"** is displayed.
- Select: **GO** (in the Rot mode window)
 - Rot mode is reloaded and rotation is restarted.
- Repeat the above test procedure for the EMERGENCY STOP button on both sides (if applicable) of the gantry control panel.

3.30.2 Checking Radiation Shut-down

SI Radiation cut-off checked (12M)



The TIMEOUT TEST checks whether the radiation OFF monitor in the generator is functioning properly. Load a scan mode with a higher number of readings than required for the respective scan time. After radiation is switched on for a period of 110% (nominal scan time plus 10%), the generator has to switch off radiation and transmit an error message.

- Enable the service functions by selecting: **Options > Local Service > Test Tools**.
- Select **Controller > Xray-Timeout**.
- Select **110%**
- Select a **mode** and press **Start**.
 - » The test is successful if the message "**Xray-Timeout test succeeded**" is displayed.
- Select the next mode.



One of each mode Topo, Sequence, and Spiral has to be tested.

3.30.3 Checking the Radiation Indicator

SI Radiation indicators checked (12M)



This test checks the function of all installed radiation "ON" indicators.

- Select any scan mode with radiation.
- Press **Start** for measurement and check whether all radiation indicators are ON.

3.30.4 Checking the Function of the Door Contact Switch

SI Door contact switch checked (12M)



Tests whether the door switch installed by the customer blocks radiation.
Omit this test if a door switch has not been installed.

- Select any scan mode with radiation.

- Press **Start** for measurement while the scan room door is open.
 - » The test is successful if the message **DOOR OPEN** is displayed.

3.30.5 Checking the Patient Positioning Aids

SI Patient positioning aids checked (12M)



This section tests the attachment of patient positioning supports to the patient table.

- Inspect the patient positioning accessories for damage (i.e. head cradle, holder, pads).
- Check the accessory bracket at the head-end of the table for damage.
- Check the lock-in mechanism on the patient positioning accessories.
 - Attach them and remove them several times from the accessory bracket at the head-end of the table.
 - The lock-in mechanism must function correctly to ensure safe and secure positioning of the accessories.



If the lock-in mechanism does not function properly, determine the cause of the problem (i.e. spring, knob, or bracket insert in table) and replace the appropriate part(s).

3.30.6 Checking the function of the Line Connection Box

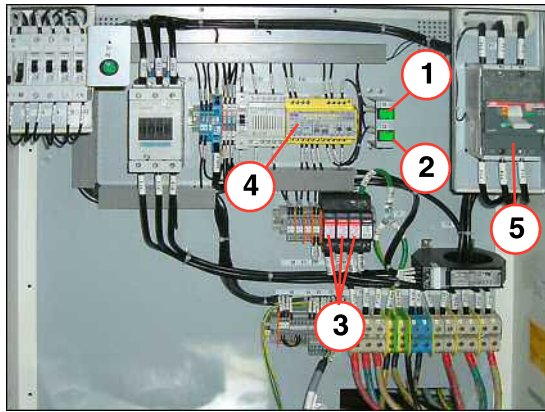
SI Function of the Line Connection Box checked (12M)



This section tests the function of the Line Connection Box.

- Switch off the system.
- Open the Line Connection Box (LCB).

Fig. 50: Line Connection Box (LCB)



- (1) Test switch S2 for the fault current switch F10
- (2) Test switch S3 for the main circuit breaker F1
- (3) 3 varistors
- (4) Fault current switch F10
- (5) Main circuit breaker F1

- Activate the test switch S2 (Pos.1) for the fault current switch F10.
 - » The test is successful if the main breaker F1 (Pos.5) has been switched off
- Switch on the main circuit breaker F1 (Pos.5).
- Activate test switch S3 (Pos.2) for the main circuit breaker F1.
 - » The test is successful if the main breaker F1 (Pos.5) has been switched off
- Switch on the main circuit breaker F1 (Pos.5).
- Check the 3 varistors (Pos.3) visually. If the window of the varistors shows red, the varistor must be replaced.

3.31 Gantry: Tilting the Gantry

Required time:	10 min.
Required tools:	n.a.
Required materials	n.a.

PM ±30 gantry tilting checked (12M)

For self-lubricating the tilt spindle:

- tilt the gantry to +30
- tilt the gantry to -30
- tilt the gantry back to the 0 position

3.32 System: Installing all Covers



Read and observe the safety information contained in the "Safety Information"(→ Safety Information / Page 20) section of this document.

SIM Covers and protective conductor cables installed (12M)

Required time:	20 min.
Required tools:	Standard tool kit
Required materials	n.a.



All metal system covers have a plug or screw connection for the protective conductor. Ensure that the protective conductor cable is connected at this point while the system covers are installed. The protective conductor cable is included with the components.

- Switch off the gantry.
- Put all cover hooks into their holders.
- Reinstall all covers.
- Switch on the gantry.

3.33 System: Image Quality Check



Read and observe the safety information contained in the "Safety information" (→ Safety Information / Page 20) section of this document.



Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained.



Before executing the measurement, system check-up should have been performed.

Q Quality check completed (12M)

Required time:	25 min.
Required tools:	Quality phantoms
Required materials	n.a.

The measurements are used for quality control and for checking the maintenance performed on the system.

The program routines may be selected via the **SERVICE/QUALITY** platform.

The **Constancy Test** automatically performs quality measurements.

- Click the **Constancy** function in the **Service/Quality** platform.

3.34 System: Cleaning

SIM Flat screen cleaned (12M)

Required time:	5 min.
Required tools:	Soft cloth; cleaning agent
Required materials	n.a.

- Clean the flat screen.
- Remove any dirt or contamination caused during maintenance.

3.35 System: Checking the System Function

Q Condition of the CT system checked (12M)

Required time:	5 min.
Required tools:	n.a.
Required materials	n.a.

- Simulate several system functions typical for the customer to ensure that the system is in good working condition before turning it over to the customer.

3.36 Maintenance Documentation



You have completed the maintenance. All work performed on the system during maintenance and the results obtained have to be documented in the "Maintenance Protocol" provided. Provide the customer with the complete protocol and, depending on organizational requirements, return a copy of it to your regional unit or field service.

3.37 End of full Maintenance

- » End of chapter



Read and observe the safety information contained in the "Safety information" (→ Safety Information / Page 20) section of this document.



Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained.

Component	Maintenance steps	Replacement interval	Estimated time
IMS	UPS Replacement	Every 2.5 years	30 min.
Total estimated time:			30 min.

4.1 Switching off the System

Required time:	2 min.
Required tools:	n.a.
Required materials	n.a.

Shut down the computers and power off the whole system and secure it against unintended switch on.

- In the ICS main user menu, select **System > End > Shutdown System** to shut down the computers.
- Switch off the power supply to the entire system and secure it against unintended switch on.

4.2 Replacing the UPS (optional)

Required time:	18 min.
Required tools	Standard tool kit
Required materials	HenDan 1500VA UPS (P/N: 10049948)



The UPS is an option part; this procedure is not necessary for the systems without UPS.



The date of replacement and the date for the next replacement have to be documented in the "Maintenance Protocol" provided.

PM UPS replaced (30M)

- Switch off the entire system.
- Disconnect the power supply for the UPS.
- Dismount the cable protection cover on the back side of the UPS.
- Remove the power supply cables between the UPS and IMS.
- Remove signal cable W53 for the ICS.
- Replace the UPS with the new one.
- Connect all new power supply cables between the UPS and IMS, and connect the new W53 signal cable for ICS as described in the Installation Instructions.
- Connect the new power supply cable for the UPS.
- Install the new cable protection cover on the back of the UPS.
- Reconnect the power supply for the UPS.

4.3 Switching on the System

Required time:	5 min.
Required tools:	n.a.
Required materials	n.a.

If the system is not in standby already, switch on the system.

- Switch on the LCB.
- Switch on the UPS for the IMS.
- Perform a system checkup.

4.4 System: Checking System Functionality

Required time:	5 min.
Required tools	n.a.
Required materials	n.a.

Q Condition of the CT system checked (24M)

- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

4.5 Maintenance Documentation



You have completed the maintenance. All work performed on the system during maintenance and the results obtained have to be documented in the "Maintenance Protocol" provided. Provide the customer with the complete protocol and, depending on organizational requirements, return a copy of it to your regional unit or field service.

4.6 End of Preventive Parts Replacement

- » End of chapter

Chapter	Change	Reason
Chapters 01, 02, 03	V05: Sliding gantry option added	ECR 614427

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